CENTRAL BANK
OF ARMENIA

THE BETTER POLICY PROJECT
GREAT THINGS ARE COMING

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Video 0 - Welcome to the Central Bank of Armenia

Note

Welcome to the Central Bank of Armenia and the **CBA Academy**! This onboarding program introduces you to the ideas, principles, and practices that guide modern monetary policy and the CBA's **Prudent Risk Management Approach (FPAS Mark II)**. Through short, focused videos, you'll explore how the CBA makes policy decisions under uncertainty—combining global lessons with Armenia's own experience. The goal is not technical mastery, but shared understanding, sound judgment, and a common language for effective decision-making.

Welcome to the Central Bank of Armenia

Hi, I'm **Vazgen Poghosyan**, Head of the Monetary Policy Department at the Central Bank of Armenia.

I want to start by welcoming you to the Central Bank of Armenia and to this special learning platform we call the **CBA Academy**.

Congratulations on your appointment!

Joining the Central Bank of Armenia is not just an honour—it's a responsibility. You're stepping into a role that helps shape the future of Armenia's economy, and we're here to ensure you're fully equipped for that task.

This onboarding journey is about more than learning the mechanics of monetary policy. It's about building a shared mindset for making decisions under uncertainty—that's what modern central banking is all about.

So, what is the **CBA Academy**?

It's our internal learning hub—a space to reflect, ask questions, and engage with the big ideas that guide our work.

We've designed this video series to help you connect the dots between **global monetary-policy thinking** and the specific framework we use here in Armenia—a framework known as **FPAS Mark II** which embodies the principles of **prudent risk management approach to monetary policy and price stability**.

The structure is simple:

1. **We begin with history**—lessons from past policy failures and successes.
2. Then we walk through the **core principles of inflation targeting**: establishing clear nominal anchors, strengthening communication, and designing resilient institutions.
3. From there, we examine the **limits of FPAS Mark I** and how **FPAS Mark II** evolved to address a more uncertain and complex world. The CBA is the first FPAS Mark II central bank in the world and it is a source of pride for the institution. Our efforts over the past few years have not gone unnoticed. For instance, after being told about the new framework at the CBA, Christopher Waller declared that the CBA was the new Reserve Bank of New Zealand meaning a central bank that is pushing the frontiers of how monetary policy analysis and communication is done. Furthermore, the National Bank of Georgia has also adopted FPAS Mark II and we see more and more central banks moving in this direction.

We'll also walk you through our **decision-making cycle**, the scenarios we use, and how staff and Board roles are designed to support **high-quality judgment**.

These are short videos—five to ten minutes each—but don't mistake brevity for simplicity. The goal isn't to give you a checklist; it's to build **intuition**—a way of thinking, a toolkit for navigating complexity.

There are no exams, no equations—just meaningful conversations backed by **practical experience** and **modern institutional thinking** from Armenia and beyond.

As you watch, I encourage you to think not only about the technical content, but also about **how we make decisions together**.

At the CBA, we take pride in being a **learning organization**.

That means we expect questions, welcome diverse views, and value reflection just as much as action.

This series is only the beginning—it's designed to get us on the same page and build a shared foundation.

But the real learning will happen in the conversations you have with staff, Board members, and—most importantly—in the **decisions you help make**.

So once again, **welcome to the CBA**.

Literature & Further Reading

- Central Bank of Armenia — Official Website
- The Better Policy Project
- Prudent Risk Management Approach to Monetary Policy (Book)
- FPAS Mark II: Avoiding Dark Corners and Eliminating the Folly in Baselines and Local Approximations

Video 1 - Lessons from the 1970s: The "Great Inflation"

Susanna Grigoryan

Recent Staff Member, Monetary Policy Department, Central Bank of Armenia

Student, Global Forecasting School

November 2025

Note

1. Overview and Historical Context

The Great Inflation of the 1970s stands as one of the defining episodes in the history of modern macroeconomics. It marked the breakdown of the post-war Keynesian consensus and forced a fundamental rethinking of how monetary policy should be conducted. Throughout the 1950s and early 1960s, policymakers believed they could fine-tune the economy, maintaining both full employment and price stability. This confidence was grounded in the apparent stability of the Phillips Curve, which suggested a reliable trade-off between inflation and unemployment.

However, the economic environment of the late 1960s and 1970s proved far more complex. Wage-setting behaviour, productivity shocks, and shifts in inflation expectations all interacted in ways that made this trade-off unstable. As inflation accelerated, the credibility of central banks deteriorated, and inflation expectations became unanchored. These developments ushered in a new era of macroeconomic instability, culminating in what became known as the Great Inflation.

2. The Phillips Curve Debate and the Natural Rate Hypothesis

At the heart of this intellectual shift were two seminal contributions: Milton Friedman's 1968 American Economic Association presidential address, "The Role of Monetary Policy," and Edmund Phelps's work on wage and price dynamics. Both argued that any apparent trade-off between inflation and unemployment was temporary. In the long run, unemployment would return to its "natural rate," determined by structural features of the labour market, while inflation would continue to rise if monetary policy remained too loose.

Friedman and Phelps introduced the idea that expectations play a central role in determining inflation outcomes. Once workers and firms anticipate higher prices, they adjust wages and contracts, accordingly, embedding inflation into the system. The result is an upward drift in inflation without any sustained employment gains. This insight marked the birth of the expectations-augmented Phillips Curve, the foundation of modern monetary policymaking.**3. Policy Failures and the Great Inflation**

The Great Inflation was not caused by a single shock but by a series of policy misjudgements that interacted with external disturbances. Expansionary fiscal and monetary policies, aimed at maintaining high employment, were pursued even as inflation pressures grew. The first oil shock of 1973–74 amplified these pressures, pushing up costs and further unsettling expectations.

Policymakers initially viewed these developments as temporary and believed inflation could be reduced gradually without recession. However, this misconception allowed the periodic oil price shocks of the 1970's to feed into higher inflation expectations, and the economy entered a vicious circle of wage-price dynamics. Attempts to restrain inflation through price and wage controls failed, and the credibility of central banks declined sharply. By the late 1970s, inflation had become deeply entrenched, with long-term expectations rising well above levels consistent with price stability.

4. The Volcker Disinflation and the Return to Credibility

It was only after a dramatic policy shift under Paul Volcker at the U.S. Federal Reserve that inflation began to fall. Volcker's approach emphasized monetary restraint and a willingness to tolerate short-term output losses to restore price stability. The resulting disinflation was painful but transformative: it demonstrated that credible policy actions could realign expectations and bring inflation under control.

Debelle and Laxton (1996) build on the Phillips curve analysis by introducing convexity and show that when inflation expectations drift higher, the costs of bringing inflation back down increase disproportionately. This explains why the disinflation in the early 1980s required such forceful action by Volcker.

The Volcker episode also illustrated a key principle of modern monetary policy, that central banks must act decisively and communicate clearly about their commitment to price stability. This experience laid the groundwork for the inflation-targeting frameworks that emerged in the 1990s and later evolved into flexible, risk-aware approaches such as FPAS Mark II.

5. Legacy for Modern Frameworks

The lessons of the Great Inflation reshaped central banking worldwide. They led to the recognition that monetary policy cannot exploit short-run trade-offs indefinitely and that credibility, transparency, and accountability are essential for maintaining price stability.

Freedman and Laxton, "Why Inflation Targeting?", explains that modern inflation targeting rests on two crucial ideas: first, there is no long-run trade-off between inflation and unemployment; and second, policymakers face a time-inconsistency problem that encourages surprise inflation without long-run benefits as initially proposed by Kydland and Prescott (1977).

In practice, this intellectual transformation gave rise to inflation targeting, a framework that emphasizes forward-looking policy, clear nominal anchors, and communication as tools for shaping expectations. As inflation targeting matured, it evolved into the Forecasting and Policy Analysis System (FPAS), which integrated modelling, scenario analysis, and structured policy discussions to support consistent, credible decision-making.

6. Armenia's Application: The Prudent Risk Management Approach

For the Central Bank of Armenia, the lessons of the Great Inflation are not abstract history. They underpin the evolution of the Prudent Risk Management Approach to Price Stability, FPAS Mark II. This framework recognizes that policymakers must operate under uncertainty and that overconfidence in a single model or baseline scenario can lead to costly mistakes.

The benefits of the new framework could already be seen prior to the official adoption of FPAS Mark II in 2025. While the new framework was being developed, the CBA was already approaching the COVID-19 pandemic era economy with risk management in mind. Prominent competing narratives in 2021 was whether inflation would be temporary or persistent. This led to the focus of using sticky and flexible price inflation as a relevant framework for analysing the economy during that time. During both the 1970s and the 2021–22 inflation surge, sticky-price components moved sharply upward, signalling persistent inflation pressures earlier than many policymakers recognized. Unfortunately, many central banks relied too heavily on mechanical nowcasting models, unreliable survey expectations, and output gap estimates that did not account for pandemic-driven supply disruptions. These errors delayed policy tightening and allowed inflationary pressures to build. Meanwhile, the CBA was one of the first central banks in the world to start tightening monetary policy to prevent inflation from getting entrenched.

FPAS Mark II moves beyond deterministic analysis toward a system that explicitly weighs risks and alternative scenarios. It builds credibility not by claiming precision, but by demonstrating prudence, acknowledging uncertainty, managing expectations, and acting consistently with a well-defined objective of price stability. This approach directly reflects the intellectual legacy of the 1970s: humility, adaptability, and a deep understanding of how expectations drive economic outcomes.

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Script

Hello everyone, I'm **Susanna Grigoryan** from the Monetary Policy Department at the Central Bank of Armenia. Welcome to this episode of the CBA Academy series.

In this video, we explore the essential lessons from the Great Inflation and how they remain directly relevant today.

A key reference for this discussion is the IMF Working Paper by Charles Freedman and Douglas Laxton, “Why Inflation Targeting?” This paper explains that modern inflation targeting rests on two crucial ideas: first, there is no long-run trade-off between inflation and unemployment; and second, policymakers face a time-inconsistency problem that encourages surprise inflation without long-run benefits.

Another important insight comes from Guy Debelle and Douglas Laxton’s 1996 paper on the convex Phillips curve. The authors show that when inflation expectations drift higher, the costs of bringing inflation back down increase disproportionately. This explains why the disinflation in the early 1980s required such forceful action.

To understand the inflation process, we must distinguish between flexible and sticky prices. Flexible prices react quickly, but they often respond to idiosyncratic shocks. Sticky prices, which make up about 70 percent of the CPI basket, adjust infrequently and reveal firms’ beliefs about inflation over the contract horizon. Sticky-price inflation provides some of the best signals about the underlying inflation process.

During both the 1970s and the 2021–22 inflation surge, sticky-price components moved sharply upward, signalling persistent inflation pressures earlier than many policymakers recognized.

Unfortunately, in 2021–22 many central banks relied too heavily on mechanical nowcasting models, unreliable survey expectations, and output gap estimates that did not account for pandemic-driven supply disruptions. These errors delayed policy tightening and allowed inflationary pressures to build.

FPAS Mark II is designed to avoid such mistakes by focusing on scenario-based analysis and embedding uncertainty directly into the policy process. It encourages economists to examine price-setting behaviour, expectations, and sectoral demand conditions—not just model-generated baselines.

Thank you for watching. I would like to thank **Douglas Laxton**, **Jared Laxton**, **Asya Kostanyan**, and **Sopio Mkervalidze** for their guidance in producing this series. See you in the next video.

See you there.

Video 2 - Monetary Targeting: Controlling Money to Control Inflation

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November 2025

Note

Monetary targeting emerged as a leading policy framework in the late 1970s and early 1980s as central banks searched for a clearer and more disciplined nominal anchor following the Great Inflation. The foundational logic came from the monetarist proposition that inflation is fundamentally the result of excessive money growth. If the central bank could restrain the growth rate of monetary aggregates to a predictable path, inflation would, over time, converge to a stable rate. This note explains the appeal of monetary targeting, the operational challenges that quickly emerged, and the lessons for good economists.

1. Logic and Appeal of Monetary Targets

The basic framework rested on three assumptions:

1. money demand was stable
2. velocity changed only gradually, and
3. output followed a predictable trend.

Under these conditions, targeting the growth rate of aggregates such as M1 or M2 seemed like a reliable method for anchoring inflation. Central banks in advanced economies, including the United States, Canada, the United Kingdom, and Germany, adopted explicit targets for money growth. Armenia, like many other countries transitioning toward modern frameworks, later experimented with similar approaches.

2. The CBA's experience with Money Aggregates

From 1994 to 2005, the Central Bank of Armenia operated under a **monetary aggregate targeting** framework. This approach was formalized with the 1996 *Law on the Central Bank of Armenia*, which established price stability as the primary policy objective, restricted direct lending to the government, mandated the use of indirect instruments, and strengthened central bank independence.

Broad money served as the **intermediate target**, while the **monetary base** operated as the **operational target**. The CBA influenced banking system liquidity and banks' investment decisions through a mix of liquidity-providing (Lombard loans, repos, open-market and FX purchases) and liquidity-absorbing (deposit auctions, repos, open-market and FX sales) instruments. The goal was to steer monetary aggregates toward planned trajectories consistent with achieving low inflation.

The regime was initially successful. Inflation, though extremely high in 1994 (1,761% year-on-year), became manageable in the second half of that year. By 1995, inflation had fallen to 32.2%, and real GDP returned to positive growth. From 1995 to 1998, inflation stabilized, and Armenia entered a low-inflation environment accompanied by rapid economic growth, including consecutive double-digit expansions from 2003 to 2007.

However, the framework soon faced structural difficulties. Until 1998, broad money (M2X)—including foreign-currency deposits—served as the nominal anchor. As dollarization increased, targeting broad money constrained growth in dram-denominated liquidity, risking a contraction in aggregate demand. To address this, the CBA began targeting dram-based aggregates in 1998, though this did not resolve inconsistencies between the operational target and the nominal anchor.

Growing volatility in money demand, the shadow circulation of U.S. dollars, rising external inflows after 2000, and a weak transmission mechanism undermined the stability of both the monetary base–broad money relationship and the broad money–inflation link. Broad money persistently deviated from targets, forcing frequent revisions and eroding the credibility of the nominal anchor. With broad money no longer a reliable anchor, the CBA faced a strategic choice between adopting an exchange-rate anchor or transitioning to inflation targeting.

3. Early Operational Difficulties

Almost immediately, policymakers encountered large and persistent deviations of monetary aggregates from their intended targets. These deviations often carried ambiguous implications for inflation. In some episodes, rapid money growth did not translate into higher inflation; in others, inflation rose even when monetary aggregates appeared to be under control. The once-stable empirical relationship between money and nominal income began to deteriorate.

4. Structural Breaks in Money Demand

Financial innovation played a central role in destabilizing previously stable money-demand functions. Deregulation in the banking sector, the introduction of interest-bearing deposits, the rise of money market mutual funds, and technological changes in payments all altered how households and firms managed liquidity. As a result, traditional aggregates like M1 and M2 ceased to track the underlying liquidity positions of the private sector. Velocity became unpredictable, and former models no longer provided reliable guidance.

A good economist recognizes these structural breaks as soon as they emerge. When a key empirical relationship becomes unstable, it cannot serve as a reliable operational foundation for policy.

5. Endogenous Money and Banking Behaviour

A deeper conceptual issue was that monetary targeting did not reflect the endogenous nature of modern money creation. When banks expand credit, deposits rise; when banks contract lending, deposits fall. These dynamics depend more on economic conditions and risk appetite than on central bank actions. Monetary aggregates therefore reflected the behaviour of private financial institutions rather than the intended stance of monetary policy. Attempts to control the quantity of money led to volatility in short-term interest rates and credit conditions, raising concerns about financial stability. Central banks were forced to choose between stabilizing interest rates or stabilizing money growth—most chose the former.

Recent research has begun filling a gap in how economists model these mechanisms. For example, DSGE models with explicit banking, balance sheet interactions, and liquidity and maturity transformations such as Benes, Kumhof, and Laxton (2014) and Mkhatrishvili (2025)—provide modern treatments of endogenous money creation. These papers are not about monetary targeting itself; rather, they offer improved analytical foundations for thinking about money, credit, and liquidity inside structural policy models.

6. Transition to Interest-Rate Instruments

As practical challenges mounted, central banks shifted from quantity-based to price-based operational frameworks. Instead of attempting to control monetary aggregates directly, policymakers began to influence economic activity and inflation by setting a Short-term policy interest rate and shaping expectations about its future path. This approach proved more effective, more flexible, and more consistent with the functioning of modern financial systems.

For Armenia, as in many other countries, experiences with monetary targeting provided valuable lessons that ultimately led to the adoption of inflation targeting frameworks supported by forecasting and policy analysis systems.

7. Lessons for Good Policy Analysis

The monetary targeting era taught economists several important lessons:

- Empirical relationships must be continuously evaluated for stability.
- Structural change can disrupt long-standing regularities.
- Monetary and credit aggregates are useful indicators but cannot serve as the sole anchor of policy.
- Good economists adapt frameworks when evidence shows that earlier assumptions no longer hold.

These insights paved the way for modern inflation-targeting regimes and the development of FPAS-type frameworks, while recent work on endogenous money creation helps rebuild the analytical foundations for thinking about money, credit, and liquidity inside those frameworks.

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Script

Hello, I'm **Susanna Grigoryan**, an economist at the **Central Bank of Armenia** and a **Level 1 student at the Global Forecasting School**. In this video, I will explain why many central banks in the 1970s and early 1980s attempted to control inflation by targeting the growth rate of money.

After the Great Inflation, policymakers wanted a clear and rule-based way to bring inflation down. Milton Friedman's idea that inflation is always and everywhere a monetary phenomenon became extremely influential. If inflation results from excessive money growth, then controlling the growth of monetary aggregates like M1 or M2 should allow the central bank to control inflation.

Central banks across the world adopted monetary targets. They announced desired paths for money growth and adjusted their policy instruments to try to keep actual money growth in line with those targets. In the early stages, this approach appeared promising. But problems quickly emerged.

Monetary aggregates often behaved unpredictably. Sometimes money growth rose sharply without producing much inflation; in other periods, inflation increased even though money growth seemed under control. These inconsistencies reflected deeper structural changes happening in the financial system.

Financial innovation — including deregulation, new types of deposits, and improved payments technology — altered how households and firms held liquid assets. Traditional measures of money no longer captured the concept reliably. At the same time, central banks discovered that they did not directly control the money supply: in modern economies, most money is created by banks through lending. When credit expanded, money grew; when credit contracted, money slowed — regardless of the central bank's intentions.

A good economist learns from these developments. When an indicator stops providing reliable information, it cannot remain the centre of a policy framework. These experiences led central banks to shift from controlling the quantity of money to influencing the price of money — the Short-term policy interest rate. This approach proved far more effective and flexible. For Armenia, as for many other countries, monetary targeting served as a transitional phase on the path toward inflation targeting supported by forecasting and policy analysis systems.

I would like to thank **Douglas Laxton**, **Jared Laxton**, **Asya Kostanyan**, and **Sopio Mkervalidze** for their guidance and support in **preparing this material.**

Video 3 - The Birth of Inflation Targeting: A New Era in Monetary Policy

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November 2025

Note

By the late 1970s and early 1980s, inflation had become deeply entrenched across both advanced and small open economies. Monetary targeting had failed as a reliable operational guide. Rapid financial innovation destabilized money demand, central banks repeatedly missed their monetary targets, and credibility deteriorated. Policymakers slowly learned a critical lesson: without anchored expectations, monetary policy cannot achieve lasting stability. Credibility—not monetary aggregates—was the scarce asset of the period. A durable monetary policy framework needed two things: (1) institutional arrangements strong enough to enforce discipline and (2) analytical tools that support coherent, forward-looking decision-making under uncertainty. New Zealand and Canada pioneered these two pillars.

1. New Zealand: The First Institutional Inflation-Targeting Regime

By the late 1980s, New Zealand faced persistent double-digit inflation, volatile expectations, and limited policy credibility. The Reserve Bank Act (1989) transformed the policy environment by establishing operational independence, a clear price-stability mandate, a Policy Targets Agreement (PTA), and direct accountability of the Governor for target outcomes. This created the first modern inflation-targeting regime that set the inflation target at 2%.

2. A Modern Rationale for a 2% Inflation Target

If you asked your grandmother what “price stability” means, she would probably tell you that *true* price stability corresponds to 0% inflation—a world where the cost of living neither rises nor falls. In principle, economists agree: zero inflation is the clean theoretical benchmark.

Yet when central banks began adopting explicit numerical inflation targets—starting with New Zealand in 1990 and soon followed by Canada—the target chosen was not zero, but 2%. This number was not revealed by economic theory; it was a pragmatic, operational choice. Policymakers wanted a rate low enough to preserve long-run price stability but high enough to avoid the well-known dangers associated with zero inflation and deflation. Over time, the 2% target became a global norm.

The reasoning rests on a few core arguments among others:

a. Downward Nominal Wage Rigidity

Workers strongly resist nominal wage cuts, even when real wages must adjust. With a small positive inflation rate, real wages can decline gradually without requiring explicit pay cuts. This improves labor-market flexibility and reduces the risk of persistent unemployment in downturns.

b. Measurement Biases in CPI

Price indices systematically overstate true inflation because they do not fully capture:

- quality improvements,
- substitution toward cheaper goods,
- the introduction of new products.

A measured 2% inflation rate may correspond to 1% or less in true inflation. Targeting 0% inflation would risk actual deflation.

c. Reducing the Risk of Deflation

Deflation is destabilizing because it:

- raises real interest rates,
- increases real debt burdens,
- depresses demand,
- tightens financial conditions.

This was the vicious cycle that perpetuated the Great Depression. A small positive target acts as a buffer, ensuring that temporary shocks do not push the economy into a deflationary spiral.

3. Canada: The First Modern Analytical Framework

However, New Zealand lacked an analytical system capable of managing the economy once inflation reached low levels. Without an FPAS, early policy decisions relied too heavily on the exchange rate and a narrow interpretation of CPI inflation and a strict period for when the inflation target should be achieved. Thus, early central bank behaviour under inflation targeting generated large and unnecessary volatility in output in order to meet its inflation objectives. Haworth, Kostanyan, and Laxton (2019) provide a detailed account of these shortcomings.

While New Zealand developed institutional credibility, the Bank of Canada developed analytical credibility. The key breakthrough was the creation of the Quarterly Projection Model (QPM) by Douglas Laxton, David Rose, and Bob Tetlow—the first forward-looking structural model used operationally for monetary policy. QPM incorporated rational expectations, a coherent transmission mechanism, and internally consistent medium-term projections, marking a decisive shift away from backward-looking econometric frameworks.

This evolution can be understood through the analogy we often use to highlight the pitfalls of relying on backward-looking and linear econometric analysis in monetary policy. The economy resembles the winding mountain road to Dilijan, not a straight U.S. interstate highway. Driving with an econometric mindset is like driving that mountain road while looking only in the rearview mirror, and assuming the road stays perfectly straight. The outcome is inevitable: you crash.

QPM, while still linear, at least allowed policymakers to keep their eyes on the road ahead. Its forward-looking structure provided a coherent way to incorporate anticipated shocks, expectations, and policy actions, laying the foundation for modern forecasting and policy analysis systems used in inflation-targeting regimes today.

QPM did not incorporate policy credibility or a loss function directly. These concepts were developed in parallel research, most importantly Laxton, Ricketts, and Rose (1993), which emphasized the importance of uncertainty, learning, and reputation. This separation ensured QPM remained tractable while credibility research informed policy decisions.

By the early 1990s, the Bank of Canada had built the world's first functioning Forecasting and Policy Analysis System. Decades later, Bernanke (2025) would highlight that advanced-economy central banks like the Bank of England and the Federal Reserve lacked precisely this type of structured FPAS framework.

4. The New Zealand Arc: From Reform to Full FPAS

- New Zealand's evolution unfolded in four stages:
- Credible disinflation (1989–1992)
- Flexibility and trade-offs (mid-1990s)
- Adoption of FPS (1997–1999)
- Modernization and macroprudential integration (2000s onward)

The RBNZ became the first central bank in the world to publish the expected path of the policy rate—an exceptional level of transparency still unmatched by many advanced economies.

The working-paper version of Hunt, Rose, and Scott (2000), presented in 1998, was the first document to use the term "Forecasting and Policy Analysis System," marking the beginning of FPAS globally.

It is no coincidence that the RBNZ's strong analytical and institutional tradition positioned it well for managing the unprecedented challenges of the COVID-19 pandemic. Building on decades of disciplined framework design and forward-looking analysis, the RBNZ was able to adopt a policy strategy that fully embraced the risk-management approach to monetary policy. During the crisis, this strategy became known as the "policy of least regrets." The idea was simple: at the depths of the pandemic, the costs of doing too little risked an economic collapse, long-term unemployment and financial instability. The RBNZ therefore opted for a strategy they knew could result in overstimulating the economy, recognizing that the extreme uncertainty, it is better to err on the side of supporting demand too much rather at the expense of a possible high inflation scenario.

5. Complementary Strengths: Institutional vs. Analytical Credibility

New Zealand pioneered institutional credibility; Canada pioneered analytical credibility. Other small open economies soon followed, notably the Czech National Bank, which implemented a full FPAS by 2003 (Coats, Laxton, and Rose 2003).

Bernanke (2024, 2025) emphasizes that many advanced-economy central banks continue to lag behind smaller open economies in the structure and transparency of their frameworks.

Armenia's FPAS Mark II integrates global best practices and builds upon them to address new challenges. It applies least-regrets analysis, forward-looking modelling, and strategic communication.

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Script

Hello everyone, I'm **Susanna Grigoryan**, an **Economist at the Central Bank of Armenia** and a **Level One student at the Global Forecasting School**. In this video, we'll explore how the modern era of monetary policy began — the birth of **inflation targeting**, a framework that transformed central banking worldwide and continues to shape our work today.

By the late 1970s and early 1980s, inflation was deeply embedded across much of the world. Monetary targeting was failing, financial innovation was breaking traditional relationships, and policymakers were struggling to anchor expectations. It became clear that credibility—not money growth—was the cornerstone of effective monetary policy.

New Zealand was the first country to tackle this challenge directly. The 1989 Reserve Bank Act introduced operational independence, a price stability mandate, and the Policy Targets Agreement, which held the Governor accountable for inflation outcomes. This was the first modern inflation-targeting regime.

At the same time, Canada was building the analytical foundations of modern monetary policy. The Quarterly Projection Model—QPM—developed by Douglas Laxton, David Rose, and Bob Tetlow, allowed policymakers to ask the right forward-looking questions: Where is the economy today? What forces are driving it? And what path for the policy rate is needed to achieve the inflation objective?

These two traditions—New Zealand's institutional credibility and Canada's analytical credibility—created the modern FPAS frameworks used around the world. New Zealand later adopted the Forecasting and Policy System, FPS, becoming the first central bank to publish the expected path of the policy rate. Even today, few advanced-economy central banks match that level of transparency.

More recently, Ben Bernanke's review of the Bank of England emphasized the importance of a comprehensive FPAS framework, scenario-based analysis, and transparent communication. These ideas were pioneered decades earlier in Canada, New Zealand, and the Czech National Bank.

Finally, Armenia's FPAS Mark II builds on all of these lessons. It is grounded in least-regrets analysis, forward-looking modelling, and transparent communication.

Thank you for watching. I would like to thank **Douglas Laxton**, **Jared Laxton**, **Asya Kostanyan** and **Sopio Mkervalidze** for their guidance in preparing this material. See you in the next episode of the CBA Academy.

See you there.

Video 4 - Principles 1 & 2 A Clear Nominal Anchor and Consistency Between Objectives

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Note

1. Why Principles Matter

The success of modern monetary policy frameworks rests on a small set of guiding principles that ensure policy is credible, consistent, and transparent. Among these, the first two are foundational: a clear nominal anchor and internal consistency among policy objectives. Without a well-understood anchor, monetary policy lacks focus and credibility. Without consistency among objectives, even a credible anchor can lose meaning over time. As Freedman and Laxton (2009) emphasize, these principles are not theoretical; they form the operational backbone of successful inflation-targeting frameworks worldwide.

2. Principle 1: A Clear Nominal Anchor

A nominal anchor defines what the central bank ultimately controls to achieve price stability. It is the fixed point that guides expectations of households, firms, and markets. In inflation-targeting regimes, that anchor is the publicly announced **point target** for inflation, supported by credible communication and consistent actions.

FPAS Mark II emphasizes a point target accompanied by a **variation band** whose purpose is only to communicate the uncertainty inherent in forecasting. The variation band is not a hard operational boundary. Attempting to force inflation to stay within the band each quarter would be counterproductive; flexible price adjustment is necessary for efficient resource allocation in the economy.

The anchor serves three essential roles:

1. Expectation Formation: It provides a benchmark for wage and price setting.

2. Policy Discipline: It constrains Short-term political or opportunistic pressures.
3. Credibility: It anchors inflation expectations, making policy more effective.

Anchored expectations reduce both inflation and output volatility, generating measurable welfare gains. Canada's experience in the early 1990s demonstrated how a transparent point target, supported by consistent policy actions, rebuilt public trust and stabilized long-term interest rates even before inflation fully converged.

3. Principle 2: Consistency Among Objectives

The second principle concerns the coherence between the inflation target and other policy goals—growth, employment, and financial stability. Freedman and Laxton (2009) argue that consistency is achieved when all objectives operate under a single overarching mandate: maintaining low, stable, and predictable inflation. Pursuing conflicting short-run goals undermines credibility and creates confusion.

The Bank of Canada's framework illustrates this principle. By keeping price stability at the core, it avoided the temptation to fine-tune short-term outcomes. Similarly, the Reserve Bank of New Zealand evolved toward a flexible framework that allowed temporary deviations during shocks without sacrificing long-run credibility.

4. Evidence from Experience

Empirical evidence shows that clear anchors and consistent objectives improve welfare. Countries with strong anchoring—such as Canada, New Zealand, the United Kingdom, and later the Czech Republic—experienced lower volatility and faster recoveries from shocks. Cross-country comparisons in Freedman and Laxton (2009) show that credibility gains were strongest where targets were explicit, communication transparent, and objectives aligned.

5. Lessons for FPAS Mark II

In FPAS Mark II, these principles define the entire forecasting and decision-making process. The point target guides scenario design, forecast discussions, and communication. Consistency ensures that messages to the public, government, and financial sector remain aligned. This disciplined yet flexible structure allows policymakers to make decisions that are credible and adaptive under uncertainty, the essence of prudent risk management.

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Script

Hello everyone, I'm **Susanna Grigoryan**, an economist at the **Central Bank of Armenia** and a **Level One student at the Global Forecasting School**. In this video, we'll turn to the first two principles that lie at the very heart of modern monetary policy frameworks: the principle of a **clear nominal anchor** and the principle of **consistency among objectives**.

These principles ensure that monetary policy is credible, coherent, and transparent. Without a clear anchor, policy loses direction. Without consistency, even a credible anchor can lose meaning. As Freedman and Laxton (2009) highlight, these principles form the operational backbone of successful inflation-targeting regimes.

Let's begin with the first principle — a clear nominal anchor. Think of it as the compass of monetary policy: the fixed point that helps everyone navigate uncertainty. In FPAS Mark II, this anchor is a **point target**, supported by a **variation band** that communicates uncertainty. The band is not something policymakers try to defend mechanically; it simply reflects the reality that inflation fluctuates as the economy adjusts.

The anchor shapes expectations, enforces discipline, and strengthens credibility. When expectations are well-anchored, inflation and output volatility fall, improving overall welfare. Canada's experience in the early 1990s shows how quickly credibility can be rebuilt when policy actions consistently support the target.

Now for the second principle — consistency among objectives. A central bank must ensure its goals work together rather than against each other. When price stability forms the foundation, other objectives such as growth, employment, and financial stability become more sustainable.

Freedman and Laxton (2009) show that successful inflation-targeting regimes align all objectives under a single overarching mandate: maintaining low, stable, and predictable inflation. Short-term trade-offs are recognized, but long-term consistency is always preserved.

Experience from Canada, New Zealand, the United Kingdom, and the Czech Republic reinforces that clear anchors and consistent objectives reduce macroeconomic volatility and build resilience.

For Armenia's FPAS Mark II framework, these principles guide the entire forecasting and decision-making process. The point target directs analysis and communication. Consistency ensures that messages to the public, government, and financial sector are aligned and credible.

To sum up, these two principles — a clear nominal anchor and consistency among objectives — are what make monetary policy effective, credible, and resilient. They help central banks stay focused on maintaining public trust and economic stability, even in uncertain times.

In our next video, we'll move on to Principles Three and Four, exploring how monetary policy interacts with fiscal policy and the real economy. Thank you for watching, and I look forward to seeing you in the next episode of the CBA Academy.

See you there.

Video 5 - Principles 3 & 4: MP and Other Policies — The Inflation–Output Trade-Off

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Note

1. Why Policy Coherence Matters

Principles 3 and 4 in Freedman and Laxton (2009) define the structural foundations of modern monetary policy frameworks. While the first two principles established the importance of a clear nominal anchor and internal consistency, these next two principles deal with how monetary policy interacts with the broader economy and how central banks organize themselves to deliver coherent, credible decisions.

Monetary policy does not operate in a vacuum. Its effectiveness depends on a consistent macroeconomic policy mix, especially coordination between monetary, fiscal, and structural policies. Equally important is having a robust operational framework that links economic analysis, policy judgment, and transparent communication. Together, these ensure that the central bank speaks with one voice and that its policy signals are coherent across time and circumstances.

2. Principle 3: Policy Coherence Across Government

The third principle emphasizes the need for coherence between monetary policy and the broader policy environment. Freedman and Laxton (2009) highlight that inflation targeting can only succeed if fiscal and structural policies support the same long-run objectives. When fiscal policy is sustainable, it reinforces the credibility of the central bank's commitment to price stability. When fiscal or exchange-rate policies conflict with the inflation objective, credibility erodes and monetary policy becomes less effective.

In practice, coherence is best achieved through cooperation, not direct coordination. Monetary and fiscal authorities should maintain a shared understanding of the economic outlook and ensure that their policies complement one another. This cooperation allows each policy area to perform well independently while contributing to overall macroeconomic stability.

However, in exceptional situations-such as during a crisis or when interest rates are at the effective lower bound-temporary coordination may be necessary to prevent deflation and sustain recovery. Large economies like the United States demonstrated this after the global financial crisis, when fiscal and monetary policies had to work hand in hand to stabilize demand. At minimum, cooperation and mutual awareness of policy stances are essential to achieve the synergies that come from well-performing policies acting together.

Canada's experience after the 1991 inflation-targeting announcement showed how credible fiscal consolidation amplified the effectiveness of monetary policy. New Zealand also demonstrated that cooperation between the government and the Reserve Bank, anchored by clear communication, was key to stabilizing expectations. Freedman and Laxton argue that coherence should never compromise independence: it is about alignment of goals, not control of instruments.

3. Principle 4: The Forecasting and Policy Analysis System (FPAS)

The fourth principle introduces the Forecasting and Policy Analysis System (FPAS) as the institutional mechanism that translates principles into practice. Freedman and Laxton describe FPAS as a disciplined process combining model-based analysis, expert judgment, and communication to guide policy decisions. It ensures that all staff work from a shared analytical framework and that policy choices are consistent with both the data and the inflation target.

FPAS consists of four main elements:

1. A quantitative model linking policy instruments to inflation and output. 2. A forecast process integrating expert judgment and scenarios. 3. A structured decision-making cycle that includes forecast rounds, policy discussions, and publication of reports. 4. A communication strategy that aligns internal analysis with external messaging.

In this way, FPAS serves as the operational backbone of modern inflation-targeting regimes. It allows central banks to manage uncertainty, update expectations in real time, and coordinate effectively across departments.

4. Lessons from Early FPAS Systems

The first generation of FPAS systems, developed at the Bank of Canada and later adapted internationally, demonstrated how analytical and institutional reforms could reinforce credibility. Adrian, Obstfeld, and Laxton (2018) show that countries adopting FPAS frameworks achieved smoother policy cycles, clearer communication, and better integration of fiscal and monetary analysis.

New Zealand's Forecasting and Policy System (FPS) and the Czech National Bank's modeling work both extended these principles to small open economies, proving that disciplined processes can enhance credibility even under uncertainty.

5. FPAS Mark II and Modern Practice

FPAS Mark II builds directly on these early systems, refining their structure to reflect new challenges-particularly nonlinearities, global spillovers, and the effective lower bound on interest rates. The modern approach emphasizes prudent risk management within the FPAS discipline: systematically identifying risks, quantifying uncertainty, and designing policies that minimize regret under different scenarios.

This evolution has allowed central banks, especially those in emerging markets, to operate with greater confidence and transparency while maintaining flexibility and consistency over time.

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Script

Hello everyone, I'm **Susanna Grigoryan**, an economist at the **Central Bank of Armenia** and a **Level One student at the Global Forecasting School**. In this video, we'll continue our exploration of the key principles that define credible and effective monetary policy frameworks. Today, we'll focus on **Principles Three and Four** from *Freedman and Laxton (2009)* — principles that deal with **policy coherence** and the **Forecasting and Policy Analysis System**, or FPAS.

While the first two principles emphasized the importance of a clear nominal anchor and internal consistency, these next two principles explain how monetary policy interacts with the broader economy, and how central banks organize themselves to make coherent and credible decisions.

Monetary policy never operates in isolation. Its effectiveness depends on a coherent macroeconomic policy mix — especially the relationship between monetary, fiscal, and structural policies. Just as important, a central bank needs a strong operational framework that links analysis, judgment, and communication. When these elements work together, the central bank speaks with one voice, and its policy signals remain consistent across time and circumstances.

Let's start with Principle Three — policy coherence across government.

Freedman and Laxton emphasize that inflation targeting can only succeed when fiscal and structural policies support the same long-term objectives. If fiscal policy is sound and sustainable, it strengthens the credibility of the central bank's commitment to price stability. But when fiscal or exchange-rate policies move in the opposite direction, credibility erodes and monetary policy becomes less effective.

In practice, coherence is achieved through cooperation, not direct coordination. Monetary and fiscal authorities must share a common understanding of the economic outlook and ensure that their policies complement one another. Each policy area can remain independent, but they should work in harmony toward overall macroeconomic stability.

In exceptional situations — such as during a financial crisis or when interest rates are constrained by the effective lower bound — temporary coordination may be needed to prevent deflation and support recovery. After the global financial crisis, large economies like the United States showed how fiscal and monetary policies could work together to stabilize demand and restore confidence.

Even outside of crises, mutual awareness and communication between authorities are essential. Coherent policies reinforce each other, while conflicting signals can undermine everything.

The experiences of Canada and New Zealand illustrate this principle clearly. In Canada, following the 1991 announcement of the inflation target, credible fiscal consolidation amplified the effectiveness of monetary policy. The government's commitment to sustainable budgets gave the Bank of Canada the space to focus entirely on achieving price stability — free from concerns about fiscal dominance.

In New Zealand, coherence took a slightly different form. The Policy Targets Agreement between the government and the Reserve Bank established shared accountability for achieving the inflation target. This institutional design ensured both sides worked toward the same long-run objective, supported by open communication and transparency.

As Freedman and Laxton point out, coherence does not mean control. It's about alignment of goals, not control of instruments. Monetary policy must remain operationally independent, but independence works best when supported by responsible fiscal and structural policies that point in the same direction.

Now let's move to Principle Four — the Forecasting and Policy Analysis System, or FPAS.

This is where the principles we've discussed so far turn into practice. FPAS is the institutional mechanism that ensures policy decisions are coherent, disciplined, and credible. It brings together models, expert judgment, structured decision-making, and clear communication — creating a unified process through which a central bank forms its view of the economy and decides on policy actions.

According to Freedman and Laxton, FPAS has four essential components.

First, a quantitative model that links policy instruments — such as the policy interest rate — to inflation, output, and other key variables.

Second, a forecast process that integrates model-based projections with expert judgment and alternative scenarios.

Third, a structured decision-making cycle, which includes forecast rounds, internal policy discussions, and publication of reports.

And **fourth**, a communication strategy that ensures consistency between internal analysis and what the central bank tells the public.

Together, these components form the operational backbone of modern inflation-targeting regimes. FPAS allows central banks to manage uncertainty, update expectations in real time, and coordinate effectively across departments — ensuring that everyone works from a shared analytical framework.

The first generation of FPAS systems, developed at the Bank of Canada, showed how disciplined processes could reinforce credibility. The system linked forecasts, models, and communication into one coherent policy cycle. Later, other central banks adapted and refined the approach. New Zealand's Forecasting and Policy System (FPS) and the Czech National Bank's modelling work extended FPAS to small open economies, demonstrating that even in uncertain environments, consistent analysis and transparent decision-making can strengthen credibility.

As Adrian, Obstfeld, and Laxton (2018) later showed, countries that adopted FPAS frameworks achieved smoother policy cycles, clearer communication, and better integration between fiscal and monetary policy.

Over time, these systems evolved into what we now call FPAS Mark II.

FPAS Mark II builds directly on these earlier systems but is designed to handle today's more complex challenges — things like nonlinear dynamics, global spillovers, and constraints from the lower bound on interest rates. The modern approach emphasizes prudent risk management within the FPAS discipline.

That means identifying risks systematically, quantifying uncertainty, and designing policies that minimize regret under a range of possible outcomes. This shift allows policymakers — especially in emerging markets — to operate with greater confidence and transparency, even when the future is uncertain.

In Armenia, this evolution is especially important. The Central Bank of Armenia's FPAS Mark II framework embodies these same principles. It combines model-based discipline with flexible, forward-looking analysis. It integrates the best international practices while adapting to the realities of a small open economy. And it ensures that every forecast, every discussion, and every policy decision contributes to a single, coherent vision for price and financial stability.

So, to summarize: Principles Three and Four remind us that monetary policy is most effective when it is coherent with broader government policies and when it operates through a disciplined analytical framework like FPAS. Coherence ensures alignment and credibility. FPAS provides the structure and process that transform analysis into action.

Together, they make monetary policy not only credible — but truly effective.

In our next video, we'll turn to Principles Five and Six, where we'll explore flexibility and transparency — the final ingredients of a modern, resilient monetary policy framework.

Thank you for watching, and I look forward to seeing you in the next episode of the CBA Academy.

See you there.

Video 5.1 - Monetary Policy Transmission Mechanisms

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Note

1. The Origins of "Long and Variable Lags"

Milton Friedman's 1968 Presidential Address introduced the famous concept of "long and variable lags." He warned that monetary policy works with unpredictable timing, meaning central banks can easily over- or under-shoot if they misjudge those delays. Even with today's advanced models, the timing and strength of policy transmission remain state-dependent, changing with fiscal conditions, wealth effects, and expectations.

2. The U.S. Experience: Delayed Tightening and the Wealth Channel

The Federal Reserve began tightening in 2022, about a year later than it should have. Rates rose by over 500 basis points in record time. Historically, this would have triggered a recession, especially with an inverted yield curve. Yet the U.S. economy avoided one.

Why? Because wealth effects offset the drag from higher interest rates. Equity prices remained high, household balance sheets were strong, and the liquidity from COVID-era fiscal support continued to circulate. Real 10-year bond yields had been deeply negative during the pandemic, fuelling asset prices and consumption.

Michael Kiley's work at the Federal Reserve shows how wealth and financial conditions can shift the timing of recessions predicted by yield curves. The lags have grown longer-not because the transmission mechanism is broken, but because other forces temporarily dominate it.

3. The Armenian Experience: Easing and Sticky Credit Rates

In contrast, Armenia has experienced the opposite problem. The central bank has cut rates sharply, but borrowing costs for households and firms remain elevated. This reflects structural features, including a concentrated banking sector, risk perceptions, and limited secondary markets. Transmission works, but slowly. The easing cycle demonstrates the asymmetry often found in small open economies: tightening bites faster than easing.

4. Why the Mechanism Appears "Broken," But Isn't

When policy effects do not appear immediately, it is tempting to assume the mechanism has failed. But the reality is more complex. Transmission operates through multiple channels: interest rates, wealth, credit, exchange rates, and expectations. At times, one channel weakens while others strengthen.

Declaring the mechanism "broken" undermines credibility. Instead, policymakers must identify which channels are active and adjust the stance or communication to maintain consistency with the inflation target. FPAS Mark II formalizes this approach: monitor, analyse, and adapt rather than overreact.

5. Policy Lessons

- The mechanism is never broken, just delayed or dampened.
- Wealth and fiscal dynamics can lengthen lags, as in the United States.
- Structural constraints can slow credit transmission, as in Armenia.
- Central banks must act decisively enough to maintain the nominal anchor.

The lesson is patience and persistence. Use the transmission mechanism you have always. If it seems slow, find out why, communicate clearly, and adjust as needed.

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Script

Hello everyone, I'm **Susanna Grigoryan**, an economist at the **Central Bank of Armenia** and a **Level One student at the Global Forecasting School**. In this extended video, we'll explore how monetary policy actually works — and why its effects often take longer to appear than people expect. This discussion brings together two important themes: the **monetary transmission mechanism**, and the idea of **long and variable lags** in policy effects.

Let's begin with the monetary transmission mechanism — the process through which central bank decisions affect inflation and output. When the central bank changes its policy rate, that action ripples through financial markets, influencing borrowing costs, exchange rates, and expectations. In other words, monetary policy works through several channels that together shape spending, investment, and inflation expectations across the economy.

There are four main channels of transmission.

The first is the interest rate channel. When the central bank raises or lowers its policy rate, it immediately changes short-term interest rates. Those shifts influence expectations about future rates, which affect long-term bond yields — for example, the ten-year government bond — and eventually the mortgage and business-loan rates that households and firms face. If we plot the ten-year bond yield and the thirty-year mortgage rate on a chart, they tend to move closely together. That relationship shows how expectations about the policy path and risk compensation determine borrowing costs throughout the economy.

The second is the exchange-rate channel. Expected interest-rate differentials across countries drive exchange-rate movements. Higher expected rates usually appreciate the domestic currency, making imports cheaper but exports less competitive. For small open economies, this is often one of the most powerful transmission routes.

The third is the credit channel. Banks' lending behaviour determines how quickly policy changes reach borrowers. If the financial system is concentrated or risk-averse, or if banks hold excess liquidity, then the transmission of policy to credit conditions can be slow and uneven.

And finally, the expectations channel — arguably the most important. When the central bank communicates clearly and consistently, it anchors inflation expectations. That credibility influences real interest rates and public confidence, ensuring that monetary policy works through both psychology and financial conditions.

In open economies like Armenia's, these channels interact with global capital flows and exchange-rate dynamics. Expectations of appreciation or depreciation combine with risk-adjusted interest-rate differentials, amplifying policy effects. Here, global financial conditions, fiscal credibility, and clear communication all shape how effectively monetary policy transmits through the economy.

Another important concept in transmission is the term and risk premium. Long-term interest rates don't move one-for-one with expected Short-term rates because investors demand compensation for uncertainty. That's the term premium — a reward for bearing interest-rate risk. In open economies, there can also be a country-risk or exchange-rate premium, reflecting investors' assessment of fiscal credibility or external vulnerability. Together, these premia explain why the ten-year government bond yield typically exceeds the policy rate by a margin that changes over time.

Now, let's turn to the second theme: why the effects of monetary policy unfold with long and variable lags. This idea goes back to Milton Friedman's 1968 Presidential Address, where he warned that monetary policy does not affect the economy instantly or in the same way every time. The transmission process is complex, influenced by wealth, expectations, and financial structure.

A good recent example comes from the United States. After the pandemic, the Federal Reserve began raising interest rates in 2022 — roughly a year later than ideal — and did so very aggressively, increasing the policy rate by more than 500 basis points. Historically, such a tightening would have signalled an impending recession, especially with the yield curve inverted. Yet, a recession never arrived.

Why? Because other forces were at work. High equity prices, strong household balance sheets, and pandemic-era savings supported consumption and delayed the impact of higher rates. During COVID, real long-term interest rates had been deeply negative, which boosted asset prices and wealth. As Michael Kiley's research shows, these wealth and liquidity conditions can lengthen the lags in transmission. The mechanism wasn't broken — it was simply slower, as other channels temporarily dominated.

Armenia's experience offers the opposite perspective. When the Central Bank eased policy, lending rates didn't fall quickly. Banks remained cautious, and the credit channel was slower to respond because of market structure and risk perception. This asymmetry is common — tightening tends to bite faster than easing — but the key point is that the transmission mechanism still works.

It's never truly broken. When one channel weakens, others can strengthen. Declaring the system broken risks undermining confidence in the nominal anchor. Instead, policymakers must identify which channels are currently active, monitor them continuously, and adjust policy instruments accordingly. That is precisely what the FPAS Mark II framework does — it provides a disciplined process for continuous monitoring, adaptation, and communication, ensuring that policy remains effective even when transmission lags change.

From this analysis, several lessons emerge for policymakers: First, expect lags to vary depending on wealth conditions, fiscal stance, and the global environment. Second, recognize asymmetry — policy tightening usually has faster effects than easing. Third, use all available channels and instruments to maintain credibility and manage expectations.

The essential message is that monetary transmission always works, but never on a fixed schedule. Patience and persistence are vital. Understanding these long and variable lags helps central banks stay disciplined, data-driven, and credible.

So, whenever someone asks whether monetary policy still works, the answer is simple: it does — it just takes time.

Thank you for watching, and I look forward to seeing you in the next episode of the CBA Academy.

See you there.

Video 6 - Principles 5 & 6: Independence, Transparency, and Accountability

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Note

1. Why analytical frameworks matter

Credible monetary policy depends not only on independence but on the discipline of an analytical framework that links forecasts, instruments, and objectives. A genuine forward-looking system has three essential ingredients:

1. Where is the economy today? A coherent and internally consistent assessment of the current macroeconomic position-output gap, inflation pressures, and financial conditions.
2. What are the underlying forces driving it? A structured analysis of demand, supply, and external shocks, showing how transmission channels operate.
3. What must be done with policy instruments to achieve the objectives? A systematic process that determines how interest rates and other instruments must adjust to keep inflation and output near target.

Any framework missing one of these three elements is incomplete. As shown in Bernanke (2024), the Bank of England never succeeded in developing this third component; hence, it never possessed a coherent forward-looking analytical framework.

2. The Bank of England's missing third ingredient

Bernanke's independent review (2024) found that the Bank's system relied heavily on judgment and an outdated model suite centred on COMPASS. Forecasts were conditioned on an exogenous or constant path for the policy rate-a practice that effectively pegged the nominal interest rate in projections. This violated the most basic principle of modern policy analysis: the policy rate must be endogenous, adjusting in response to the forecast itself.

As Milton Friedman (1968) warned in his Presidential Address, pegging the nominal rate is inherently destabilizing. If inflation turns out higher than assumed, the real rate falls, stimulating demand and further raising inflation. Thus, an exogenous rate path magnifies, rather than dampens, instability.

In a proper analytical framework, the forecast model determines the interest-rate path needed to achieve the inflation target, consistent with the central bank's reaction function. Its absence at the Bank of England-and similarly at other large, bureaucratic central banks-meant that forecasts could not reliably guide policy or communicate a credible story to the public.

3. Contrast with first-generation FPAS frameworks

Central banks such as the Reserve Bank of New Zealand, the National Bank of Georgia, and the Central Bank of Armenia built what we describe as Mark I Forecasting and Policy Analysis Systems (FPAS). These frameworks explicitly integrate:

- A structural model linking key economic relationships
- A forecast process that endogenizes the policy instrument
- Scenario analysis for prudent risk management
- A communication system that transparently explains policy choices

In these institutions, the forecast and the policy decision are two sides of the same coin. The analytical framework ensures every published scenario is consistent with the central bank's objectives and its understanding of transmission mechanisms.

4. The state of the art: The Prudent Risk Management Approach

The Prudent Risk Management Approach to Price Stability (2023) defines the state-of-the- art analytical framework. It integrates forecasting, policy, and communication into a unified decision system. Policy is derived from an explicit loss function balancing inflation and output stabilization under uncertainty, with scenario analysis to quantify risks.

This eliminates the artificial separation between analysis and decision, ensuring that the central bank's instruments, objectives, and communications are fully aligned. Compared with this benchmark, the Bank of England's structure (Bernanke 2024) was not merely outdated-it was conceptually incomplete. It failed to answer the essential question: What must be done with our instruments to achieve our objectives?

5. Implications for institutional integrity

Independence protects a central bank from political pressure, but independence without an analytical framework breeds confusion and weak accountability. Transparency and accountability arise when decisions can be traced through a disciplined process linking evidence, analysis, and action.

The experiences of the RBNZ, NBG, and CBA show that smaller, more agile institutions can achieve world-class standards when they combine operational independence with analytical coherence and clear communication.

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Script

Hello everyone, I'm **Susanna Grigoryan**, an economist at the **Central Bank of Armenia** and a **Level One student at the Global Forecasting School**. In this video, we'll look at why independence, transparency, and accountability are the cornerstones of credible monetary policy.

Credible monetary policy depends not only on independence but on the discipline of an analytical framework that links forecasts, instruments, and objectives. A genuine forward-looking system has three essential ingredients. First, where is the economy today? It requires a coherent and internally consistent assessment of the current macroeconomic position—output gap, inflation pressures, and financial conditions. Second, what are the underlying forces driving it? That means a structured analysis of demand, supply, and external shocks, showing how transmission channels operate. And third, what must be done with policy instruments to achieve the objectives? It demands a systematic process that determines how interest rates and other instruments must adjust to keep inflation and output near target.

Any framework missing one of these three elements is incomplete. The Bank of England, as shown in **Bernanke (2024)**, never succeeded in developing this third component; hence, it never possessed a coherent forward-looking analytical framework.

Bernanke's independent review in 2024 found that the Bank's system relied heavily on judgment and an outdated model suite centred on COMPASS. Forecasts were conditioned on an exogenous or constant path for the policy rate—a practice that effectively pegged the nominal interest rate in the projections. This violated the most basic principle of modern policy analysis: the policy rate must be endogenous, adjusting in response to the forecast itself.

As **Milton Friedman (1968)** warned in his Presidential Address, pegging the nominal rate is inherently destabilizing. If inflation turns out higher than assumed, the real rate falls, stimulating demand and further raising inflation. In this way, an exogenous interest-rate path magnifies, rather than dampens, instability.

In a proper analytical framework, the forecast model determines the interest-rate path needed to achieve the inflation target, consistent with the central bank's reaction function. This link between objectives and instruments is what allows the policy process to be internally coherent and forward-looking. Its absence at the Bank of England—and similarly at other large, bureaucratic central banks—meant that forecasts could not reliably guide policy or communicate a credible story to the public.

By contrast, central banks such as the **Reserve Bank of New Zealand**, the **National Bank of Georgia**, and the **Central Bank of Armenia** built what we describe as Mark I Forecasting and Policy Analysis Systems, or FPAS. These frameworks explicitly integrate a structural model linking the economy's key relationships, a forecast process that endogenizes the policy instrument, scenario analysis for prudent risk management, and a communication system that explains policy choices transparently. In these institutions, the forecast and the policy decision are two sides of the same coin. The analytical framework ensures that every published scenario is consistent with the central bank's objectives and its best understanding of transmission mechanisms.

The **Prudent Risk Management Approach to Price Stability (2023)** defines the state-of-the-art analytical framework. It integrates the forecasting, policy, and communication functions into a unified decision system. Policy is derived from an explicit loss function that balances inflation and output stabilization under uncertainty, using scenario analysis to quantify risks. This approach eliminates the artificial separation between analysis and decision, ensuring that the central bank's instruments, objectives, and communications are fully aligned. Compared with this benchmark, the Bank of England's structure revealed by **Bernanke (2024)** was not merely out-of-date—it was conceptually incomplete. It failed to answer the third essential question: what must be done with our instruments to achieve our objectives?

Independence protects the central bank from political pressure, but independence without an analytical framework breeds confusion and weak accountability. Transparency and accountability arise when decisions can be traced through a disciplined process linking evidence, analysis, and action. The experiences of the **RBNZ**, **NBG**, and **CBA** show that smaller, more agile institutions can achieve world-class standards when they combine operational independence with analytical coherence and clear communication.

Independence gives us the power to act. Transparency and accountability give us the legitimacy to lead. Together, they form the foundation of institutional integrity and public trust.

Thank you for watching, and I would like to acknowledge **Douglas Laxton, Jared Laxton, Asya Kostanyan, and Sopio Mkervalidze** at the **Global Forecasting School** for their guidance and support in producing this video.

See you there.

Video 7 - When the World Changed: The Limits of FPAS Mark I

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November 2025

Note

1. FPAS Mark I and Its Strengths

FPAS Mark I frameworks, inspired by the work of Freedman and Laxton (2009), were built on a clear and simple logic. They combined macroeconomic models, expert judgment, and structured policy processes to ensure internal consistency between forecasts and decisions. The approach worked effectively in relatively stable times. Baseline forecasts provided a coherent narrative about where the economy was heading, and policy recommendations could be evaluated against this most-likely outlook. For two decades, this approach supported credible inflation-targeting regimes across advanced and emerging- market economies.

2. The Three Layers of Uncertainty

The world that FPAS Mark I was designed for assumed a stable structure and modest, mean- reverting shocks. But beginning in the late 2010s, three forms of uncertainty began to dominate policymaking:

- Structural uncertainty - permanent changes to the global economy caused by COVID-related shocks, geopolitical realignments, and deglobalization.
- Model uncertainty - the erosion of stable relationships such as the Phillips Curve, with nonlinearities, supply constraints, and credibility effects.
- Policy-credibility uncertainty - the risk that inflation expectations could de-anchor when shocks persist and communication loses clarity.

These layers of uncertainty made it impossible to rely on a single, deterministic baseline as the guide for decisions. The very foundation of FPAS Mark I-the idea that the central bank could identify and optimize around one "most-likely" scenario-became untenable.

3. The Limits of Baseline Determinism

In the classical FPAS structure, the baseline scenario represented the best estimate of the future. Policy recommendations were derived from this single narrative. But the experience of 2020–2023 demonstrated how fragile such determinism can be. When multiple shocks interact-pandemics, supply bottlenecks, energy crises-there is no single trajectory the economy will follow. Instead, central banks face a range of possible outcomes, many of them shaped by behavioural feedback, credibility shifts, and nonlinear responses.

A deterministic baseline not only misrepresents reality; it also biases communication. By projecting a false sense of precision, it discourages policymakers from discussing risks openly. As shown in The Prudent Risk Management Approach to Price Stability (Central Bank of Armenia, 2025), effective communication under uncertainty requires transparency about risks-not just point forecasts.

4. Lessons from 2020–2023

The pandemic and subsequent global supply shocks, as well as geopolitical tensions, exposed the operational limits of FPAS Mark I. Across both advanced and emerging economies, baseline-driven forecasts failed to anticipate the persistence of inflationary pressures. In some cases, this was due to structural changes in labour markets and global trade; in others, it reflected insufficient integration between judgment and model-based analysis. Central banks that clung to deterministic baselines were repeatedly forced to revise forecasts upward and backtrack on forward guidance. Confidence in the forecasting process-and by extension, in monetary policy itself-suffered.

The experience reaffirmed that policy under uncertainty cannot be based on a single optimal path. Instead, it must recognize and manage the spectrum of possible risks that define the new environment.

5. Implications – Toward FPAS Mark II

The recognition that uncertainty is not a temporary disturbance, but a structural condition led to a profound rethinking of the FPAS architecture. The Central Bank of Armenia, drawing on lessons from global experience, developed the Prudent Risk Management Approach to Price Stability-FPAS Mark II. This new system embeds risk analysis within every stage of the policy process, from forecasting to communication. Its guiding principle is prudence: minimizing potential regret rather than optimizing a single baseline scenario. By embracing uncertainty, FPAS Mark II transforms it from a source of vulnerability into a foundation for robust policymaking.

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Script

Hello everyone, I'm **Susanna Grigoryan**, an economist at the **Central Bank of Armenia** and a **Level One student at the Global Forecasting School**. In this video, we'll explore how the world changed—and why traditional forecasting systems could no longer keep up.

The evolution of modern monetary policy frameworks has always reflected the world around them. During the 1990s and early 2000s, the first generation of Forecasting and Policy Analysis Systems—FPAS Mark I—served central banks remarkably well. They provided structure, discipline, and coherence in policy formulation and communication. But by the early 2020s, the global environment had changed beyond recognition. Pandemic disruptions, geopolitical conflicts, and deglobalization revealed deep vulnerabilities in these deterministic systems. The experience underscored a profound truth: the world had become too uncertain for single-baseline policymaking.

FPAS Mark I frameworks were built on a clear and simple logic. They combined macroeconomic models, expert judgment, and structured policy processes to ensure internal consistency between forecasts and decisions. The approach worked effectively in relatively stable times. Baseline forecasts provided a coherent narrative about where the economy could be heading, and policy recommendations could be evaluated against this most-likely outlook. For two decades, this approach supported credible inflation-targeting regimes across advanced and emerging-market economies.

The world that FPAS Mark I was designed for assumed a stable structure and modest, mean-reverting shocks. But beginning in the late 2010s, three forms of uncertainty began to dominate policymaking: structural uncertainty, meaning permanent changes to the global economy caused by pandemics, geopolitical realignments, and deglobalization; model uncertainty, meaning the erosion of relatively stable relationships such as the Phillips Curve, with nonlinearities, supply constraints, and credibility effects; and policy-credibility uncertainty, meaning the risk that inflation expectations could de-anchor when shocks persist and communication loses clarity. These layers of uncertainty made it impossible to rely on a single, deterministic baseline as the guide for decisions. The very foundation of FPAS Mark I—the idea that the central bank could identify, optimize and communicate around one “most-likely” scenario—became untenable.

In the classical FPAS structure, the baseline scenario represented the central bank’s best and “most plausible” estimate of the future, and policy recommendations were derived from this single narrative. But the experience of 2020 to 2023 demonstrated how fragile such determinism can be. When multiple shocks interact—pandemics, supply bottlenecks, energy crises, wars—there is no single trajectory the economy will follow and it is impossible to derive a most “plausible scenario.”. Instead, central banks face a range of possible outcomes, many of them shaped by behavioral feedbacks, credibility shifts, and nonlinear responses. A single deterministic baseline not only misrepresents reality; it also biases communication. By projecting a false sense of precision, it discourages policymakers from discussing risks openly. As shown in The Prudent Risk Management Approach to Price Stability (Central Bank of Armenia, 2025), effective communication under uncertainty requires transparency about risks, not just point baseline forecasts.

The pandemic and subsequent global supply shocks exposed the operational limits of FPAS Mark I. Across both advanced and emerging economies, baseline-driven forecasts failed to anticipate the persistence of inflationary pressures. In some cases, this was due to structural changes in labour markets and global trade; in others, it reflected insufficient integration between judgment and model-based analysis. Central banks that clung to deterministic baselines were repeatedly forced to revise forecasts upward and backtrack on forward guidance. Confidence in the forecasting process—and by extension, in monetary policy itself—suffered. The experience reaffirmed that policy under uncertainty cannot be based on a single optimal path. Instead, it must recognize and manage the spectrum of possible risks that define the new environment.

The recognition that uncertainty is not a temporary disturbance, but a structural condition led to a profound rethinking of the FPAS architecture. The **Central Bank of Armenia**, drawing on lessons from global experience, developed the **Prudent Risk Management Approach to Price Stability—FPAS Mark II**. This new system embeds risk analysis within every stage of the policy process, from forecasting to communication. Its guiding principle is prudence: minimizing potential regret rather than optimizing a single scenario. By embracing uncertainty, FPAS Mark II transforms it from a source of vulnerability into a foundation for robust policymaking.

For many years, FPAS Mark I served central banks well. It offered structure and discipline, helping us explain our policy decisions through a single baseline forecast. But the world we live in today is not the same as it was twenty years ago. The pandemic, global supply shocks, and geopolitical tensions showed us that uncertainty is no longer the exception—it's the rule.

FPAS Mark I was built on the idea that we could predict the most likely path for the economy and optimize policy around it. That worked relatively well when the world was stable. But once multiple shocks hit simultaneously, the idea of a single baseline became unrealistic. We needed a system that could manage uncertainty, not ignore it. That's why the Central Bank of Armenia developed the **Prudent Risk Management Approach to Price Stability**, or FPAS Mark II. It's a framework designed for a world of uncertainty, where policy is not about optimizing a single baseline forecast but about managing risks and minimizing regret. By embracing this mindset, policymaking institutions can make policy decisions that remain credible, flexible, and forward-looking—no matter how unpredictable the world becomes.

Thank you for watching, and I would like to acknowledge **Douglas Laxton, Jared Laxton, Asya Kostanyan, and Sopio Mkervalidze** at the **Global Forecasting School** for their guidance and support in producing this video.

See you there.

Video 8 - FPAS Mark II: The Least-Regrets Mindset

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Note

1. Historical Lineage

The Reserve Bank of New Zealand launched the first inflation-targeting regime in 1990 and, by 1998, published a comprehensive manual outlining the Forecasting and Policy Analysis System (FPAS). This framework set the international standard for disciplined, forward-looking monetary policy. Many central banks - including Canada, Chile, the Czech Republic, and Israel - adopted the FPAS principles, integrating them into their inflation-targeting systems.

But over time, several large, advanced-economy institutions - notably the Federal Reserve, the Bank of England, the European Central Bank, the Bank of Japan, and the Bank of Korea failed to develop comparable systems. They remained reliant on fragmented forecasting methods and judgmental processes without a coherent analytical framework linking models, scenarios, and policy decisions.

After the global financial crisis, the RBNZ began using the term "least regrets" to describe a cautious approach to balancing risks between over-tightening and under-tightening policy. However, this remained a communications device, not a transformation of the policy framework itself. No changes were made to the structure of its FPAS or to the way risks were managed and formally integrated into the decision process.

2. The Analytical Leap

The Central Bank of Armenia's innovation was to take the "least regrets" concept and operationalize it through FPAS Mark II. Mark II is designed for a world characterized by deep uncertainty - where the nature, persistence, and interaction of shocks are impossible to model into one optimal baseline projection.

Instead of optimizing policy around a single baseline scenario, Mark II embeds risk analysis into every stage of the policy process. Uncertainty is not treated as noise but as the core feature of the environment in which policy must operate.

Mark II combines several key elements:

1. Modelling framework capable of handling nonlinearities in the economy and endogenous policy credibility.
2. A decision framework based on management and least-regrets analysis.
3. A communication strategy emphasizing transparency about possible policy responses under uncertainty rather than false precision.
4. Continuous development of human capital in a dynamic learning environment.

This integration transforms monetary policy from a static optimization problem into a dynamic process of risk management and learning.

3. Institutional Comparisons

The difference between Armenia and other central banks is not one of resources but of institutional mindset. Large institutions - such as the Fed, ECB, and Bank of England - continue to operate without integrating analysis, decision-making, and communication. They have not yet built even the first generation of FPAS that New Zealand pioneered in the 1990s.

In contrast, Armenia developed the second generation - FPAS Mark II - by combining structured projection and policy analisys with the flexibility of prudent risk management. This was a leap forward in institutional design, not an incremental upgrade.

Waller's observation that "Armenia has become the new New Zealand" reflects the growing recognition that innovation in monetary frameworks increasingly comes from smaller, agile institutions.

4. The Least-Regrets Principle in Practice

In FPAS Mark II, every policy round begins with a structured scenario matrix. Staff with close and continuous involvement of Board members jointly evaluates risks and uncertainties in the economy. At the beginning of each policymaking round, Board members independently prepare one-page initial macroeconomic risk assessments, where they list out the main risks and uncertainties (subject to the 3R constraint) that they deem to be the most pressing and worthy of internal discussion and external communication. The staff's role in the meantime is, based on deliberations with Board members, preparing high-quality and reasonable risk scenarios.

In FPAS Mark II framework, at least 3 types of scenarios are prepared: a scenario that is based on market's expectations, scenarios that result in the policy rate following a higher path than what is priced in markets (Case A) and scenarios where the policy rate would follow a lower path than what is currently priced into markets (Case B). Additionally, when appropriate, tail risk scenarios including Case X (policy rate path significantly higher than market expectations) and Case Y (significantly below market expectations) could be included, which would need to be managed to avoid throwing the economy into dark corners.

For each scenario, the analysis evaluates the costs of acting too soon versus too late. The objective is to choose the policy stance that minimizes potential regret across all possible futures.

Mistakes are asymmetric:

- Tightening too early may unnecessarily suppress demand and employment.
- Tightening too late may allow inflation expectations to de-anchor.

The art of policymaking under FPAS Mark II is selecting the stance that minimises policy error across all plausible outcomes - not optimizing for a single point forecast.

5. Implications for Credibility and Communication

The least-regrets framework transforms communication. Instead of presenting a single baseline projection, the central bank explains:

- how it manages uncertainty,
- how it evaluates risks,
- how it would respond if conditions differed from expectations.

This shift makes transparency meaningful: the public understands both what the central bank expects and how it plans to act across different environments e.g. if certain risks materialize or economic conditions change.

FPAS Mark II completes the intellectual journey that began in New Zealand after the global financial crisis. For the first time, the least-regrets idea has moved from language to institutional reality. Christopher Waller, while at a conference in June 2025 at the Bank of Korea, recognized this achievement when he remarked that once he understood what had been built in Armenia, he realized that Armenia had "become the new New Zealand." This statement captured a generational shift — from New Zealand as the pioneer of the first FPAS in 1990 to Armenia as the pioneer of the second generation, FPAS Mark II.

CENTRAL BANK OF ARMENIA

THE BETTER POLICY PROJECT

GREAT THINGS ARE COMING

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Script

Hello everyone, I'm Susanna Grigoryan, an economist at the Central Bank of Armenia and a Level One student at the Global Forecasting School. In this video, we'll explore FPAS Mark II — the least-regrets mindset — and how it represents a decisive intellectual leap in modern monetary policy.

The evolution from FPAS Mark I to FPAS Mark II represents a decisive intellectual leap in modern monetary policy. The term "least regrets" was first popularized by the Reserve Bank of New Zealand after the global financial crisis, but it remained a rhetorical concept rather than an operational framework. In contrast, the Central Bank of Armenia transformed the idea into a full analytical system. Christopher Waller, while at the "Structural Shifts and Monetary Policy" conference held at Bank of Korea from June 1-3 recognized this achievement when he remarked that once he understood what had been built in Armenia, he realized that Armenia had "become the new New Zealand." This statement captured a generational shift — from New Zealand as the pioneer of the first FPAS in 1990 to Armenia as the pioneer of the second generation, FPAS Mark II.

The Reserve Bank of New Zealand launched the first inflation-targeting regime in 1990 and, by 1998, published a comprehensive manual outlining the Forecasting and Policy Analysis System, or FPAS. This framework set the international standard for disciplined, forward-looking monetary policy. Many central banks, including Canada, Chile, the Czech Republic, and Israel, adopted the FPAS principles, integrating them into their inflation-targeting systems. But as time passed, several large, advanced-economy institutions — notably the Federal Reserve, the Bank of England, the European Central Bank, the Bank of Japan, and the Bank of Korea — failed to develop comparable systems. They remained reliant on fragmented forecasting methods and judgmental processes without a coherent analytical framework linking models, scenarios, and policy decisions.

After the global financial crisis, the Reserve Bank of New Zealand began using the term "least regrets" to describe its cautious approach to balancing risks between over-tightening and under-tightening policy. However, this remained a communications device, not a transformation of the policy framework itself. No changes were made to the structure of its FPAS or to the way risks were formally integrated into the decision process.

The Central Bank of Armenia's innovation was to take the "least regrets" concept and operationalize it through FPAS Mark II. Mark II is designed for a world characterized by deep uncertainty — where the nature, persistence, and interaction of shocks are impossible to impossible to model into one optimal baseline projection. Instead of optimizing policy around a single baseline scenario, Mark II embeds risk analysis into every stage of the policy process. This means that uncertainty is not treated as noise but as the core feature of the environment in which policy must operate.

The analytical foundation of Mark II combines several key elements. First, a modelling framework capable of handling nonlinearities in the economy and endogenous policy credibility. Second, a decision framework based on risk management and least-regrets analysis. Third, a communication strategy that emphasizes transparency about uncertainty rather than false precision. Forth, realization that human capital is the key and need to be continuously developed in a dynamic learning environment. This integration transforms monetary policy from a static optimization problem into a dynamic process of risk management and learning.

The difference between Armenia and other central banks is not one of resources but of institutional mindset. Large, established institutions such as the Fed, the ECB, and the Bank of England continue to operate without fully integrating analysis, decision-making, and communication. They have yet to build even the first generation of FPAS that New Zealand pioneered in the 1990s. In contrast, Armenia has already developed the second generation directly — FPAS Mark II — by combining the discipline of structured forecasting with the flexibility of prudent risk management. This was a leap forward in institutional design, not an incremental upgrade.

Waller's observation that "Armenia has become the new New Zealand" is significant because it reflects the recognition that innovation in monetary policy now comes from smaller, more agile institutions. Just as New Zealand redefined central banking three decades ago by introducing inflation targeting, Armenia is now redefining it again by introducing a full risk-management-based policy framework.

In FPAS Mark II, every policy round begins with a structured scenario matrix. Staff with close and continuous involvement of Board members jointly evaluates risks and uncertainties in the economy. At the beginning of each policymaking round, Board members independently prepare one-page initial macroeconomic risk assessments, where they list out the main risks and uncertainties (subject to the 3R constraint) that they deem to be the most pressing and worthy of internal discussion and external communication. The staff's role in the meantime is, based on deliberations with Board members, preparing high-quality and reasonable risk scenarios.

In FPAS Mark II framework, at least 3 types of scenarios are prepared: a scenario that is based on market's expectations, scenarios that result in the policy rate following a higher path than what is priced in markets (Case A) and scenarios where the policy rate would follow a lower path than what is currently priced into markets (Case B). Additionally, when appropriate, tail risk scenarios including Case X (policy rate path significantly higher than market expectations) and Case Y (significantly below market expectations) could be included, which would need to be managed to avoid throwing the economy into dark corners.

The least-regrets mindset recognizes that policy mistakes are asymmetric. For example, tightening too early could unnecessarily suppress demand and employment, while tightening too late could allow inflation expectations to de-anchor. The art of policymaking under FPAS Mark II is to find the policy stance that performs acceptably across all plausible outcomes, not to optimize one point forecast. This approach provides stability and resilience, even when the economy is hit by multiple interacting shocks.

The least-regrets framework also transforms communication. Instead of presenting a single baseline projection, the central bank explains how it manages uncertainty and evaluates trade-offs. This makes transparency meaningful: the public sees not only what the central bank expects, but how it would respond if the world evolves differently. In a world where credibility depends on humility and adaptability, this approach represents a major advance in communication strategy.

FPAS Mark II thus completes the intellectual journey that began in New Zealand in 1990. It integrates the lessons of three decades of inflation targeting and provides the blueprint for the next generation of credible monetary policy frameworks. For the first time, the least-regrets idea has moved from language to institutional reality.

When the Reserve Bank of New Zealand pioneered inflation targeting in 1990, it changed the way the world thought about monetary policy. By 1998, it had published a full manual describing how a Forecasting and Policy Analysis System — or FPAS — should work. That became the blueprint for what we now call FPAS Mark I. For many years, this system worked well in both advanced and emerging economies. But over time, most of the large central banks — the Fed, the Bank of England, the ECB, the Bank of Japan, and the Bank of Korea — never developed their own versions of this framework. They relied on judgment and fragmented models rather than a coherent system for linking analysis, decisions, and communication.

After the global financial crisis, New Zealand began using the phrase “least regrets” to describe a cautious approach to policy. It was a good phrase, but nothing changed in the underlying framework. There was still only one baseline forecast, and uncertainty was treated as something outside the model.

Here in Armenia, we took that same idea and made it real. FPAS Mark II is the first complete system that turns least-regrets thinking into a disciplined analytical framework. It accepts that uncertainty is not temporary — it's the permanent condition of policymaking. Instead of trying to predict a single future, we prepare for several plausible ones. We evaluate the costs of acting too soon or too late and then choose the policy that we would least regret if the world turns out differently than we expect.

This least-regrets mindset changes everything. It means we build our forecasts, our policy discussions, and our communication around uncertainty — not around false precision.

The result is a system that combines independence, accountability, transparency, and analytical rigor in a single process. The central bank communicates the risks it sees in the economy as well as the scenarios to deal with them. It's not about predicting the future perfectly, but about managing risk responsibly and having least regrets by minimizing policy error

So, when people say Armenia has become the new New Zealand, it's not just a compliment — it's a recognition that the frontier of central banking has moved. Independence gives us the authority to act. Prudence and transparency give us the legitimacy to lead.

Thank you for watching, and I would like to acknowledge **Douglas Laxton**, **Jared Laxton**, **Asya Kostanyan**, and **Sopio Mkervalidze** at the **Global Forecasting School** for their guidance and support in producing this video.

Video 9 - Scenario-Based Forward Guidance

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November 2025

Note

1. The Concept of a Policy Round

A policy round under FPAS Mark II is a structured decision-making system that integrates four key stages: assessment, scenario development, policy decision, and communication. Unlike traditional frameworks where forecasting and policy are separated, the Central Bank of Armenia (CBA) treats these stages as a unified process.

A distinctive innovation is that the policy round is not linear but a continuous loop: once the Board makes a decision, the system transitions directly into evaluation and preparation for the next round. This institutional feedback structure transforms the CBA into a dynamic learning organization, improving judgment, and enhancing communication with each iteration.

At the core of the policy round is the creation of

All FPAS Mark II scenarios should answer the following questions:

1. Where is the economy now?
2. What are the underlying forces driving the economy?
3. How should policy adjust its tools to achieve its targets.

2. Stage One: Where is the economy today?

Every projection and policy analysis cycle begins with a fundamental question: where is the economy now? Economists use their expertise to interpret current economic conditions, considering how different developments fit together and what they reveal about the underlying state of the economy. This requires judgment—understanding how the economy is behaving, and assessing which forces are likely to matter most.

A key part of this stage is evaluating whether the risks and developments observed are temporary fluctuations or signs of deeper, more persistent shifts in the economy. Economists translate these assessments into narratives that explain what is happening and why. These narratives are not mechanical outputs; they reflect informed reasoning about the economy's structure, behaviour, and likely trajectory.

As new information emerges, these narratives evolve. The most important requirement is that they remain conceptually consistent across different areas of the economy—growth, inflation, labour markets, external conditions, and expectations. Once established, these coherent narratives become the starting point for constructing future risk scenarios.

The Staff are encouraged to challenge the status quo about the underlying features in the economy. This typically means hypothesizing about alternative views about key unobserved variables such as potential output, the NAIRU, the country risk premium, the neutral interest rate, and inflation expectations both domestically and in foreign economies. The implications of this analysis can change our views about where the economy is today. The narratives evolve as new information emerges, and staff continuously revisit their initial assessments—refining existing narratives or developing new ones when economic conditions shift. Although high-frequency indicators provide timely signals about developments in inflation, labour markets, external conditions, credit activity, and expectations, their value lies in how economists interpret them, not in the data alone. These signals help inform real-time judgments about the economy's direction, but it is the staff's economic reasoning—rather than any model or statistical tool—that ultimately shapes the diagnosis of current conditions and the understanding of underlying forces. For example, in 2022, when Russia invaded Ukraine, the general consensus at the time among the Board was that this was going to have negative implications on the Armenian economy. Meanwhile, there was at least one board member at the time, who had a completely different interpretation of the shock, who said that it would ultimately benefit the Armenian economy i.e. higher potential output. In this instance, the narratives about the state of the economy could be very different from each other moving forward depending on one's views of potential output which would have significant implications on estimates of the output gap.

Staff meet with the Board on a regular basis to update them on current economic conditions and the evolving narratives. These weekly briefings—held without fail every Wednesday afternoon—ensure that the assessment of risks, uncertainties, and underlying forces remains a continuous and ongoing conversation, rather than something compressed into the days immediately preceding a policy decision. This steady rhythm of discussion keeps the Board closely connected to the evolving economic landscape throughout the entire policy cycle.

3. Stage Two: Scenario Development - What are the underlying forces?

At this stage, staff begin constructing the Mark II illustrative scenarios. They receive each Board member's *Initial Macroeconomic Risk Assessment*—the one-pager that outlines the member's interpretation of the key forces shaping the economy, the risks and shocks they consider most relevant, and the adverse developments they believe warrant the greatest attention. These one-pagers provide the starting point for translating individual perspectives into a coherent set of scenarios that reflect the full range of concerns within the Board.

Staff construct a set of internally consistent scenarios: a market reference, a Case A where the path of the policy rate is above the market reference scenario, and a Case B where the path of the policy rate is below the market reference scenario. These scenarios incorporate shocks relevant to Armenia's economic environment, including commodity prices, external demand, financial conditions, and domestic risks.

Each scenario includes a narrative about what drives economy under that path. The Staff uses suite of models that provide quantitative projections of narratives and risk scenarios. However, the projections are ultimately expert-driven with extensive analysis and judgement to support their plausibility. In FPAS Mark II, scenarios are not alternatives to choose from but tools to evaluate risks and trade-offs. They help assessing where the balance of risks lies and how those risks could shift.

4. Stage Three: Policy Decision - The Least-Regrets Evaluation

Against the backdrop of entire policy round's lively discussions and debates of the multiple scenarios and risks, along with the submission of the final illustrative market reference, Case A and Case B scenarios, Board members would take a vote on the policy decision. The Board's decision would be announced in tandem with issuance of the Monetary Policy Report, which would clearly communicate through a narrative approach the Board's decision with reference to the scenarios and ingredients considered by policymakers to be most relevant to the current situation and its uncertainties.

As part of the decision-making process, board members would make submissions that include:

- the policy action they propose;
- how that action connects to what they believe may happen in the future and its ensuing policy implications.

- and their commitment to changing course if new information arises.

5. Stage Four: Communication and Accountability

The final stage of the policy round transforms internal deliberations into clear, public-facing communication. The CBA publishes the policy decision together with an accompanying Press Release, and the Monetary Policy Report (MPR) is released on the same day. The announcement is followed by a Governor-led press conference with members of the media, which is live-streamed to ensure the broader public receives a timely and transparent explanation of the decision and the reasoning behind it.

The CBA publishes MPR on the same day as policy decision. The CBA emphasizes scenario analysis rather than a single deterministic forecast. Communication explains:

- the economic context,
- the rationale for the decision,
- the risks evaluated, and
- the structure of the decision process.

This transparency enhances accountability. Clear communication also strengthens transmission: when markets understand how policy responds to risks, expectations adjust more effectively.

7. Conclusion - Institutional Discipline as Prudence

The policy round is the operational backbone of FPAS Mark II. It embeds prudence, transparency, and accountability into the very structure of decision-making. Rather than claiming certainty, the CBA demonstrates credibility by showing how uncertainty is managed.

This framework has positioned Armenia as an international model for agile, modern inflation- targeting institutions. The combination of independence, structure, and disciplined learning defines what credible policy looks like in today's uncertain macroeconomic environment.

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Script

Hello everyone, I'm Susanna Grigoryan, an economist at the Central Bank of Armenia and a Level One student at the Global Forecasting School. In this video, we'll go inside FPAS Mark II to understand how the anatomy of a policy round works and how it has made the Central Bank of Armenia a global benchmark in operationalizing the principles of FPAS Mark II.

Unlike many institutions that treat forecasting, policy formulation, and communication as separate activities, the Central Bank of Armenia integrates them into a unified and disciplined process known as the policy round. This cyclical process ensures that each stage—from data assessment to communication—feeds back into the next, creating a continuous loop of analysis, decision, and learning. The policy round is not simply a meeting schedule; it is the institutional architecture through which prudence, transparency, and accountability are embedded into every monetary policy decision.

At its core, a policy round under FPAS Mark II is a structured decision-making cycle linking four essential stages: assessment, scenario development, policy decision, and communication. Each stage involves collaboration across departments—Research, Statistics, Financial Stability, and Communication—and each contributes uniquely to the final decision. This process ensures that the central bank acts with coherence, internal accountability, and clear alignment between its analytical framework and policy objectives. The CBA's innovation lies in treating the policy round as a continuous loop rather than a linear sequence. After each decision, the process transitions immediately into evaluation and feedback, preparing the institution for the next round. This dynamic structure transforms the central bank from a forecasting institution into a learning organization—one that refines its models, judgment, and communication strategy with every iteration.

The first stage of the policy round is devoted to developing a well-grounded assessment of the current economic environment. Although the Staff monitors a wide range of timely indicators—including inflation dynamics, labor market conditions, credit developments, external trends, and expectations—the analytical focus lies in interpreting what these signals collectively reveal about underlying economic forces. The emphasis is on economic reasoning: evaluating whether observed developments reflect temporary fluctuations or more persistent structural shifts, understanding how domestic and global influences interact, and assessing how these interactions shape near-term macroeconomic conditions.

This stage is not a mechanical reading of recent data, but a disciplined exercise in economic judgment. Staff synthesize the various strands of information into internally consistent narratives that explain the state of the economy and the forces driving it. This diagnostic exercise answers the first essential question of the policy framework: where is the economy today, and what explains its current trajectory?

At this stage, staff begin constructing the Mark II illustrative scenarios. They draw on each Board member's Initial Macroeconomic Risk Assessment—the one-pager that outlines the member's interpretation of the forces shaping the economy, the risks and shocks they view as most relevant, and the adverse developments they believe warrant particular attention. These one-pagers serve as the starting point for translating individual perspectives into a coherent set of scenarios that reflects the full range of concerns within the Board.

In the next step, staff then develop a set of internally consistent scenarios: a market reference scenario, a Case A scenario in which the policy rate path lies above the market reference, and a Case B scenario in which it lies below. Additional illustrative scenarios may be constructed when the Board identifies a wider set of risks. These scenarios incorporate shocks relevant to Armenia's economic environment, including changes in commodity prices, external demand, financial conditions, domestic pressures, and broader geopolitical uncertainties.

Each scenario is accompanied by a narrative that explains the economic forces driving that path—how inflation, activity, and expectations would evolve under the assumed conditions, and why. The staff use the full suite of models to quantify these narratives and to ensure that each scenario describes a macroeconomically coherent adjustment back toward equilibrium. However, the projections are ultimately expert-driven: they rely on economic expertise, sectoral knowledge, and careful interpretation to ensure that each scenario is plausible and analytically sound.

In FPAS Mark II, scenarios are not alternatives to select among but tools for evaluating risks and policy trade-offs. They help the Board assess where the balance of risks lies, how those risks might shift, and what that implies for a prudent policy stance.

With the full policy round's discussions completed—and after the final market reference, Case A, and Case B illustrative scenarios have been submitted—the Board turns to the policy decision itself. Each member evaluates the scenarios, the associated risks, and the uncertainties emphasized throughout the round, and then submits their proposed course of action. These submissions explain:

- the policy action the member supports,
- how this action fits with their judgment about the way the economy is likely to evolve and the risks they consider most salient, and
- their readiness to adjust the policy stance should incoming information alter the economic landscape.

The final stage of the policy round translates complex internal deliberations into clear public communication. The Monetary Policy Report serves as the main vehicle for this transparency. Its narrative explains the economic context, the reasoning behind the decision, and the risks considered. Unlike traditional reports that focus on a single baseline forecast, the CBA's Monetary Policy Report highlights scenario analysis and conveys how the Board weighed competing risks. This approach fosters accountability by allowing external observers—markets, academics, and the public—to evaluate the coherence between the central bank's stated framework and its actions. The clarity of this communication also enhances the effectiveness of policy transmission, as market participants adjust expectations in response to the central bank's explanation of how uncertainty is managed.

The policy round is the operational heartbeat of FPAS Mark II. It institutionalizes prudence through structure, integrates transparency into every step of the process, and transforms uncertainty from a vulnerability into a manageable feature of policy. The CBA's approach demonstrates that credibility does not come from claiming certainty but from building systems that handle uncertainty systematically. In this way, Armenia has set a new global standard for how small, agile institutions can lead the next generation of inflation-targeting frameworks.

Thank you for watching, and I would like to acknowledge **Douglas Laxton, Jared Laxton, Asya Kostanyan, and Sopio Mkervalidze** at the **Global Forecasting School** for their guidance and support in producing this video.

See you there.

Video 10 - Building Scenarios: The 3R Principle

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Note

1. The 3R Principle - Related, Relevant, Realistic

When considering how the economy may evolve, policymakers face an almost limitless set of potential scenarios. In periods of elevated uncertainty, it is always possible to imagine an ever-growing list of developments. But an unrestricted set of scenarios is not helpful. It burdens policymakers and external audiences with unnecessary possibilities and does little to support effective management of risks. Scenario selection must therefore follow a disciplined approach to ensure that only those scenarios that genuinely matter for monetary policy are examined.

FPAS Mark II employs a structured filter—the **Three-R (3R) approach**—to determine which scenarios are worth considering in policy deliberations. A scenario must be:

1. Related: It must be firmly connected to the current conjuncture and consistent with the range of plausible interpretations of the data. Initial conditions are never known with precision, and uncertainty about the state of the economy is itself a key motivation for constructing scenarios. A related scenario draws directly from this uncertainty and reflects a meaningful interpretation of the forces operating in the economy today.

2. Realistic: It must describe an evolution of the economy that *could reasonably occur*, even if it is not the most likely outcome. The aim is not to predict the future with precision but to establish credible reference points that capture how the economy might respond under alternative conditions. A realistic scenario maps out developments that are worth thinking through in advance—plausible paths that help policymakers understand the potential range of outcomes.

3. Relevant for Monetary Policy: It must address risks or developments that would materially affect the conduct of monetary policy over the medium term. If a shock could push the economy toward a harmful outcome—or a “dark corner”—in the event of a policy mistake, it merits consideration. By contrast, scenarios that are not rooted in current conditions or that have little bearing on price stability do not contribute to sound policy deliberation.

Applying the 3R criteria ensures that the Board concentrates its attention on risks and uncertainties with direct implications for achieving the price stability objective. Hypothetical or extreme shocks that are not connected to the current state of the economy—for example, an unlikely catastrophic event—may be conceivable but provide no guidance for today’s policy stance.

The process for identifying candidate scenarios is inclusive. It draws on the perspectives of Board members (expressed in their Initial Macroeconomic Risk Assessments), the staff’s ongoing analysis of the economy, and market participants’ views as captured in surveys and financial indicators. The staff, under the direction of the Head of the Monetary Policy Department, screens these inputs to ensure scenarios meet the required criteria. Over time, continued dialogue and training encourage all participants to naturally converge toward proposing scenarios that satisfy the 3R principles.

2. Why Communication Defines Credibility

In FPAS Mark II, credibility is no longer measured by the central bank’s ability to predict short-term inflation with precision. Instead, credibility emerges from transparency: how clearly the central bank explains uncertainty, how consistently it applies its analytical framework, and how responsibly it manages risks.

Communication is not supplementary-it’s integral. In a world shaped by persistent shocks and nonlinear dynamics, credibility must be earned continuously through clarity, discipline, and communication about uncertainty.

3. The Credibility Cycle

Credibility under FPAS Mark II is dynamic. It evolves with every policy round through three reinforcing pillars:

- Consistency - Aligning the analytical framework, policy stance, and communication.
- Transparency - Explaining uncertainty, trade-offs, and the rationale for decisions.
- Accountability - Reviewing past projections and publicly reporting results.

These three pillars create a "credibility cycle." Each round begins with internal analytical consistency, continues through transparent decision-making and communication, and ends with a public evaluation of outcomes. This process then repeats, strengthened by learning and feedback.

4. How FPAS Mark II Structures Communication

Communication in FPAS Mark II is tightly aligned with analysis and policy. The Monetary Policy Report (MPR) is no longer a passive summary-it is an explanatory document that shows how the Board assessed risks and reached its decision.

Each MPR presents:

- a market reference scenario,
- Case A-type scenario —reflecting narratives about economic and financial developments that would require a higher interest rate path than what is currently priced in financial markets,
- Case B-type scenario, which would require a lower policy rate path in the medium-term in order to reach price stability.

paired with a detailed discussion of risk asymmetries around each. It explains how the least- regrets principle guided the Board's choice and how the costs of alternative actions were weighed.

Press conferences and interviews then reinforce this narrative by emphasizing prudence, learning, and humility rather than false precision.

5. Endogenous Credibility: How Communication Reinforces Policy

Credibility in FPAS Mark II is endogenous-shaped by what the central bank does and how it explains its actions. When the public understands the logic behind decisions, expectations adjust smoothly and with less volatility. Transparent communication reduces uncertainty about the reaction function, making monetary transmission more efficient.

In contrast, inconsistent or vague communication erodes credibility, raising doubts about whether decisions are grounded in analysis or driven by discretion. By embedding structured communication within the analytical and policy framework, FPAS Mark II makes credibility self-reinforcing.

6. Institutional Design: Internal Accountability

Internal communication is as important as external communication. After each policy round, the CBA conducts a Credibility Review: a formal meeting where staff and Board assess:

- forecast performance,
- market interpretations,
- clarity of communication, and
- alignment between internal analysis and external messaging.

Findings from these reviews directly shape the next MPR and the next communication cycle. This disciplined process eliminates gaps between internal views and public messages - a common source of credibility failures in other central banks.

7. Comparative Perspective

Many large central banks-such as the Federal Reserve, Bank of England, and ECB, treat communication as a post-decision output rather than as part of the policy process itself. This has resulted in inconsistent messaging and misaligned forward guidance during periods of high uncertainty.

In contrast, FPAS Mark II integrates communication from the very beginning of the policy round. The CBA's approach emphasizes structure, transparency, and humility, treating uncertainty not as a weakness but as an inherent feature of the world that must be managed openly.

8. Conclusion – Credibility Through Clarity

The Three-R principle ensures that only policy-relevant scenarios enter the FPAS Mark II process. A scenario must be *related* to the current economic context, reflecting plausible interpretations of today's conditions; it must be *realistic*, describing developments that could credibly unfold even if they are not the most likely; and it must be *relevant*, capturing risks that have direct implications for monetary policy and the medium-term inflation outlook. By applying these criteria, FPAS Mark II narrows a limitless universe of hypothetical paths to a concise set of scenarios that genuinely matter for informed policy deliberation.

The essence of FPAS Mark II communication is turning uncertainty into credibility. By explaining how risks are managed-rather than pretending they can be eliminated-the CBA strengthens transparency, accountability, and trust.

Credibility becomes not a reputation inherited from the past, but a continuously earned outcome of prudent and structured communication.

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Script

Hello everyone, I'm Susanna Grigoryan, an economist at the Central Bank of Armenia and a Level One student at the Global Forecasting School. In this video, we begin with one of the foundations of FPAS Mark II: the **3R Principle** for scenario selection – *Related, Realistic, and Relevant*. This principle not only guides how we think about uncertainty, but also plays a central role in how the CBA builds credibility under FPAS Mark II.

In this framework, credibility is not built by predicting short-term inflation with great precision. It is earned by showing the public that we manage uncertainty in a disciplined, structured, and responsible manner. The Three-R Principle is the first step in that discipline. FPAS Mark II does not consider an unlimited universe of hypothetical scenarios. Instead, a scenario must be **related** to current economic conditions, **realistic** in the sense that it could credibly unfold, and **relevant** for monetary policy and the medium-term inflation outlook. This filter narrows countless possibilities down to the specific scenarios that truly matter for informed policymaking.

Once the Board and staff identify scenarios that meet the Three-R criteria, the CBA constructs the **Market Reference Scenario**, a **Case A scenario** that would require a higher policy rate path, a **Case B scenario** that would require a lower path, and additional illustrative scenarios when needed. These are not alternatives the Board chooses between. Instead, they form a structured map of risks that helps policymakers understand the range of plausible outcomes and the trade-offs they must manage.

This scenario-based structure leads directly into the broader risk-management approach of FPAS Mark II. Under this system, credibility is not about offering a single, deterministic forecast. It is about showing how uncertainty is interpreted, how risks are weighed, and how policy decisions are grounded in a transparent and internally consistent framework.

Communication plays a central role in this process. In FPAS Mark II, communication is not an afterthought added at the end of the decision. It is an integral part of the analytical and policy cycle. The way the CBA communicates *follows the same structure* that guides internal analysis: the scenarios considered, the risks evaluated, and the least-regrets principle applied when reaching a decision.

Credibility under FPAS Mark II is therefore **dynamic**. It evolves through a cycle with three reinforcing pillars. First, **consistency**, meaning alignment between the analytical framework, the policy stance, and the messages conveyed to the public. Second, **transparency**, which requires clear explanations of uncertainty, trade-offs, and the reasoning behind decisions. Third, **accountability**, which means evaluating past projections and decisions and reporting the results publicly.

Together, these pillars form a self-reinforcing credibility cycle. Each policy round begins with internal analytical discipline, continues through transparent communication, and ends with a public review of outcomes. The lessons learned feed directly into the next round.

The Monetary Policy Report is the main document that makes this process visible. It no longer presents a single forecast. Instead, it provides a narrative explanation of how the Board assessed risks through the Market Reference, Case A, and Case B scenarios, and how the least-regrets principle guided the policy decision. By focusing on narratives and risks, rather than point predictions, the MPR shows the public *how* decisions were reached, not just *what* those decisions were.

Press conferences and interviews reinforce this structure by emphasizing prudence, learning, and humility rather than certainty. The aim is not to claim knowledge of the future, but to show that the central bank is prepared for a range of plausible outcomes and that policy decisions are grounded in coherent logic.

A key insight of FPAS Mark II is that **credibility is endogenous** – it is created by what the central bank does and how clearly it explains its decisions. When communication is structured and transparent, expectations adjust smoothly, monetary transmission becomes more effective, and trust strengthens over time. When communication is inconsistent or vague, credibility weakens, and uncertainty increases.

Internal communication is just as important as external communication. After each policy round, the CBA holds a **Credibility Review** to evaluate forecast performance, market interpretation, and the effectiveness of communication. The conclusions from this review directly shape the next MPR and the next communication cycle, ensuring that internal and external messages remain aligned.

Many large central banks still treat communication as something prepared after the decision has already been made. This often leads to inconsistencies, especially during periods of high uncertainty. FPAS Mark II takes a different approach. Communication is embedded from the start of the policy round, integrated with analysis and decision-making, and designed to reflect institutional humility and transparency.

The essence of FPAS Mark II communication is to make **uncertainty credible**. By explaining how risks and trade-offs are managed, rather than **pretending they can be eliminated,** the CBA strengthens transparency, accountability, and ultimately credibility. In this framework, credibility is not a static reputation but a continuously earned result of disciplined communication and consistent decision-making.

In the past, credibility was often judged by forecast accuracy. Today, in an environment shaped by significant uncertainty, credibility is built by showing how decisions are made. At the CBA, communication is an integral part of the analytical process itself. Each policy round ends with a clear explanation of what we have learned, what risks matter most, and how we balanced those risks in making the final decision. This approach allows the public to trust not that we know the future, but that we manage uncertainty with discipline.

FPAS Mark II shows that credibility and communication are inseparable. Transparency creates understanding. Understanding builds trust. And trust anchors expectations—the ultimate aim of monetary policy.

Thank you for watching. I would like to acknowledge Douglas Laxton, Jared Laxton, Asya Kostanyan, and Sopio Mkervalidze at the Global Forecasting School for their guidance and support in producing this video.

See you there.

Video 11 - Changing Roles: Board and Staff under FPAS Mark II

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November 2025

Note

1. Analytical Context

FPAS Mark II was developed in response to rising structural uncertainty - including pandemics, geopolitical shocks, and nonlinear global dynamics. Traditional baseline-focused methods proved too rigid in an environment where shocks were persistent and interacted in complex ways. FPAS Mark II integrates risk-based policymaking within a coherent scenario framework, enabling decision-makers to compare alternative worlds rather than rely on a single deterministic forecast.

2. Board as Decision-Makers under Uncertainty

Under FPAS Mark II, Board members are decision-makers operating under uncertainty. Their responsibility is to assess the distribution of risks around the inflation target. Using the least-regrets mindset, they evaluate policy-relevant scenarios and consider how shocks could evolve depending on global conditions, domestic transmission channels, and behavioural responses.

The Board's goal is not to approve a single forecast but to choose the policy stance that minimizes potential losses if the world unfolds differently than expected. This approach anchors credibility: even when outcomes deviate from projections, the decision is justified based on prudent risk management.

3. Staff as Facilitators of Structured Analysis

The shift to FPAS Mark II significantly transforms staff roles. Rather than defending a baseline, staff act as facilitators of structured analysis. Their responsibilities include:

- Designing coherent scenarios using the 3R principle - Related, Relevant, Realistic.
- Quantifying these scenarios with nonlinear models incorporating endogenous credibility and convex Phillips curve dynamics.
- Presenting uncertainties and trade-offs clearly in the Executive Briefing.

Staff are no longer advocates of a single view but architects of the analytical framework, helping the Board understand trade-offs without prescribing decisions.

4. Institutional Learning and Human Capital

FPAS Mark II embeds continuous learning into the policy process. After each policy round, staff and Board conduct structured reviews of forecast errors, communication outcomes, and shifts in economic relationships. This feedback loop strengthens human capital, improves models, and enhances institutional credibility.

As described in *The Prudent Risk Management Approach to Price Stability* (Central Bank of Armenia, 2025), FPAS Mark II transforms forecasting from a one-off analysis into a dynamic, iterative process. Each round builds institutional memory and deepens analytical capacity.

5. Policy Implications

This evolution in roles yields three major benefits:

- **Transparency** - separating analysis from judgment clarifies responsibilities.
- **Adaptability** - scenario-based evaluation allows policy to adjust to uncertainty.
- **Credibility** - prudence becomes embedded in every stage of the policy cycle.

By aligning Board judgment with structured staff analysis, FPAS Mark II ensures disciplined, forward-looking policymaking.

The effectiveness of FPAS Mark II depends not only on its procedures and analytical structure, but also on the people who operate within it. The framework requires a culture of professionalism, disciplined reasoning, and constructive debate—both among the Board and within the Monetary Policy Department. While culture is often difficult to define, strong leadership and well-designed institutional arrangements can shape how the organization interprets information, deliberates, and makes decisions. Under FPAS Mark II, human capital becomes a central pillar of the framework's practical implementation.

6. Role of the Board

Formally, the Board's mandate is straightforward: to ensure that price stability is achieved over the medium term. In practice, this task becomes considerably more complex in an environment of uncertainty and frequent shocks. FPAS Mark II equips the Board with a systematic risk-management framework, built on principles such as least-regrets, scenario analysis, and structured assessment of uncertainties. But for these principles to translate into effective decisions, the Board's deliberative culture must evolve alongside the technical framework.

A key shift under FPAS Mark II is the move toward a culture of **adversarial collaboration**. The framework is designed to avoid groupthink and to encourage Board members to contribute independent perspectives. Each policymaking round begins with Board members submitting *Initial Macroeconomic Risk Assessments*: concise one-page documents outlining the risks, uncertainties, and scenarios they judge most important—subject to the 3R criteria. These submissions form the basis for the master scenario taxonomy used throughout the round. This arrangement ensures that individual concerns are represented and that differing views are incorporated into both internal deliberations and external communications.

The least-regrets framework reinforces this independence. When casting their votes, Board members may reach the same policy conclusion for entirely different reasons—for example, one member may emphasize domestic demand risks while another focuses on global financial conditions. FPAS Mark II explicitly allows for this diversity of reasoning. What matters is not uniformity of views, but that each member makes a disciplined assessment of which risks they would most regret materializing.

Accountability strengthens this system. Every Board member provides a written explanation of their vote, describing the risks that shaped their decision and how their preferred action helps manage those risks. These explanations are published alongside the policy decision. They give the public insight into differing views within the Board and reinforce a culture in which decisions must be justified in clear, coherent terms.

Taken together, these arrangements create a deliberative environment in which opposing views can coexist productively. Adversarial collaboration does not seek consensus for its own sake; rather, it allows diverse beliefs to inform a more robust risk-management process. This culture aligns with FPAS Mark II's recognition that uncertainty is pervasive and that high-quality decisions emerge from structured debate, not from forced agreement.

7. Role of the Staff

While the Board is responsible for policy decisions, the Staff's role is to support the Board with rigorous, objective economic analysis. Their task is not to advocate for a particular policy stance but to ensure that the Board has access to comprehensive, well-reasoned assessments of the economy and the uncertainties surrounding it. In many ways, the staff enable the Board to function effectively within a culture of adversarial collaboration.

High-quality analysis requires approaching every issue through the lens of uncertainty. Whether the topic is the output gap, labor market dynamics, inflation pressures, or external risks, staff must explore multiple hypotheses, consider alternative interpretations of the data, and articulate several coherent narratives. The aim is not to determine the single "correct" interpretation but to present the full range of plausible views that policymakers must consider.

This approach demands professionalism, critical thinking analytical rigor, and disciplined reasoning. Staff must be able to construct and explain coherent scenarios on different sides of an issue, applying the same level of care and consistency to each interpretation. This mode of analysis reflects the principles of "lateral thinking" described by Edward de Bono: separating judgment from evaluation, and deliberately examining issues from several angles before drawing conclusions.

A simple example illustrates this approach. When the unemployment rate rose from 13.1% to 15.5% in early 2024, the staff did not present a single explanation. Instead, they developed multiple hypotheses: one suggesting weakening labor market conditions, another showing that the rise was driven by labor supply dynamics rather than job loss, and a third reflecting a mix of these factors. Each hypothesis was examined using available data—including growth in registered employment and wage developments—and presented without judgment about the appropriate policy reaction. This allowed the Board to consider several scenarios and understand the risks associated with each one.

This division of responsibilities—staff providing a broad, unbiased analytical foundation and the Board making judgments—serves several important purposes. It reinforces the central role of uncertainty in the policy process, ensures that the Board is exposed to a full range of perspectives, and supports lively but structured deliberation. Ultimately, it enables decisions to be made according to the least-regrets and risk-management principles at the heart of FPAS Mark II.

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Script

Hello everyone, I'm Susanna Grigoryan, an economist at the Central Bank of Armenia and a Level One student at the Global Forecasting School. Welcome back to the CBA Academy Learning Hub. In this session, we'll look at how the roles of the Board and the staff evolve under FPAS Mark II — the Prudent Risk Management Approach to Price Stability.

Under FPAS Mark I, most central banks relied on a single baseline forecast. Policy discussions revolved around one expected path for inflation and output, and decisions were evaluated largely against that forecast. As uncertainty in the global economy deepened — from pandemics to geopolitical shocks to nonlinear global dynamics — this deterministic approach became too narrow. FPAS Mark II replaces the single-path mindset with a structured scenario framework. The focus shifts from trying to predict one future to managing risks across many plausible futures.

In this environment, Board members are true decision-makers under uncertainty. Their responsibility is not to endorse a forecast but to assess the distribution of risks around the economy. They weigh policy-relevant scenarios — the Market Reference, Case A, Case B, and others that meet the 3R criteria of being Related, Realistic, and Relevant. Using the least-regrets mindset, they consider how global developments, domestic transmission channels, and behavioral responses could evolve. Their task is to choose the policy stance that would minimize losses if the world unfolds differently than expected. This approach anchors credibility because decisions are justified by prudent risk management, not by the illusion of precision.

At the same time, the role of the staff has changed fundamentally. Under FPAS Mark I, staff tended to defend a baseline projection. Under FPAS Mark II, they act as facilitators of structured analysis. Their job is to design coherent scenarios using the 3R Principle, quantify them with nonlinear models that incorporate features like endogenous credibility and convex Phillips-curve dynamics, and present the trade-offs clearly in the Executive Briefing. Staff are no longer advocates of one view; they are the architects of the analytical framework that helps the Board understand the full range of possibilities.

Models remain essential tools, but their purpose has shifted. The value of a model is judged by whether it helps answer three core questions: Where is the economy today? What forces are driving its trajectory? And what does policy need to do to achieve the price-stability objective? Models that require heavy fixes or that generate unstable statistical outputs play a limited role. The focus is on models that reinforce human capital, improve understanding of transmission mechanisms, and help structure uncertainty — not on mechanical forecasting exercises.

The relationship between the Board and the staff becomes more collaborative and dynamic. Board members bring judgment, experience, and policy perspective. Staff bring structure, quantification, and clarity. Together, they develop a shared language for interpreting risks and evaluating policy choices. Policy rounds, scenario briefings, and continuous exchanges ensure that analysis and judgment evolve together.

This redistribution of roles did not happen in a vacuum. FPAS Mark II emerged from recognizing that modern central banking must adapt to persistent uncertainty. By embedding risk-based policymaking within a scenario-driven structure, FPAS Mark II enables decision-makers to compare alternative worlds rather than rely on a single deterministic path.

For Board members, this means evaluating the full distribution of risks. Each policymaking round begins with the Initial Macroeconomic Risk Assessments — the one-page documents where Board members list the scenarios and uncertainties they believe matter most. These submissions shape the master taxonomy of scenarios for the round and ensure that each member's concerns are taken seriously. When voting on the policy rate, Board members may reach the same decision for different reasons, reflecting their own assessments of the risks they would most regret. These individual explanations, published alongside the policy decision, reinforce accountability and allow the public to understand how different perspectives contributed to the outcome.

For staff, the analytical responsibility is equally significant. Their role is to prepare the scenario taxonomy, test multiple hypotheses on each key issue, and present a balanced assessment of the uncertainties. They must analyze data from several angles — without bias, without defending a preferred interpretation, and without implying a single correct answer. This approach reflects the logic of lateral thinking: examining issues comprehensively, acknowledging uncertainty, and helping the Board consider the full range of plausible outcomes.

FPAS Mark II also transforms forecasting into an ongoing institutional learning process. Every policy round includes a structured review of forecast performance, market interpretations, and communication outcomes. These reviews feed directly into the next round, strengthening models, improving communication, and deepening institutional expertise. Over time, this continuous learning process builds credibility — not through perfect predictions, but through demonstrated discipline and openness.

The evolution in roles under FPAS Mark II delivers three essential benefits. It increases transparency by clarifying the distinction between analysis and judgment. It strengthens adaptability by allowing policy **to respond to uncertainty rather than resist it.** And it supports credibility by embedding prudence, structure, and accountability in every stage of the policy cycle.

In short, FPAS Mark II transforms the institution. Board members make decisions under uncertainty. Staff provide the analytical structure that enables robust deliberation. Models support human judgment without substituting for it. Together, this forms the core of the Prudent Risk Management Approach to Price Stability — a living framework where every policy round reinforces institutional credibility.

Thank you for watching, and I would like to acknowledge Douglas Laxton, Jared Laxton, Asya Kostanyan, and Sopio Mkervalidze at the Global Forecasting School for their guidance and support in producing this video.

See you there.

CENTRAL BANK
OF ARMENIA

THE BETTER POLICY PROJECT
GREAT THINGS ARE COMING

Video 12 - Policy-Making Cycle and Policy-Relevant Analysis

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November 2025

Note

1. The 29-Day Policy Cycle: From Event-Based to Continuous Learning

The Prudent Risk Management Approach to Price Stability also known as FPAS Mark II, represents a shift toward recognizing uncertainty as the natural state of the world. Forecasting is no longer an act of prediction but an act of disciplined learning under uncertainty. This perspective shapes the 29-day policy cycle and the new concept of policy relevant analysis.

FPAS Mark II replaces event-driven policy routines with a structured 29-day cycle that integrates analysis, judgment, and communication. Each stage of the cycle has clearly assigned responsibilities:

- **T–29: Kick-Off:** A concise staff briefing introduces the Board to the current economic drivers and key uncertainties. This is followed by an open and robust discussion among Board members about the major concerns and risks shaping the upcoming policy round.
- **Next 2 Weeks: Initial Risk Assessments:** Each Board member prepares a one-page narrative outlining the risks, uncertainties, and issues they believe must be managed during the current policymaking round. These submissions form the foundation for the taxonomy of scenarios.
- **T–15: Scenarios Meeting:** Board members present their Initial Risk Assessments, which are consolidated into the overall Taxonomy of Scenarios. Staff then present the illustrative Case A and Case B scenario narratives that will be quantitatively modeled for the round.
- **T: Decision Day:** Staff deliver the final presentation on current and underlying economic conditions. This sets the stage for Board deliberations and, ultimately, the policy decision.

As the FPAS Mark II manual emphasizes, regularity builds credibility while dynamism sustains learning. The cycle forms a continuous loop in which each round informs the next.

Kick-Off Meeting (T-29)

The round begins with a Kick-Off Meeting where staff deliver a concise briefing on key economic drivers, underlying trends, and major sources of uncertainty. This is followed by an open, wide-ranging discussion among Board members on the risks and issues they believe warrant attention — such as global financial vulnerabilities, domestic demand pressures, or the possibility of drifting toward high- or low-inflation traps.

The goal is not to review data mechanically but to surface the deeper concerns shaping each member's thinking. From the staff's perspective, the Kick-Off Meeting consolidates analysis and discussion from weekly briefings leading up to the round. In extraordinary circumstances, the meeting may also be used to recalibrate the work plan for the round.

Developing Scenario Narratives (Next 10 Days)

Following the Kick-Off Meeting, Board members begin developing the core ideas that will feed into the Case A and Case B scenarios. Case A reflects situations requiring a policy rate path above market expectations, while Case B reflects circumstances requiring a lower path. Board members are not asked to construct full scenarios; instead, they outline the essential narratives, shocks, and assumptions they believe should guide scenario development.

These ideas are set out in their **Initial Macroeconomic Risk Assessments**, or "one-pagers." Each one-pager provides a narrative on the Board member's key risks and uncertainties, subject to the **3R Principle**: scenarios must be *Related* to current conditions, *Realistic* enough to merit consideration, and *Relevant* for monetary policy.

Importantly, these documents are not binding. Board members are encouraged to revise their thinking as the round unfolds — this flexibility is an intentional feature of FPAS Mark II.

Process for Preparing One-Pagers (T-29 to T-18)

Board members work with two assigned staff members ("Board Coordinators") who support them throughout the process:

1. **Initial Brainstorming Session (1 hour):** Board members share preliminary bullet points on their key risks. Coordinators facilitate discussion, ensure alignment with

the 3R criteria, ask clarifying questions, and reference insights from the Kick-Off Meeting and analytical briefings.

2. **Drafting the One-Pager (2–4 hours):** Coordinators prepare the first draft based on the brainstorming session.
3. **Iterations (1–2 hours)**: The Board member and coordinators refine the draft over email during the second week.
4. **Submission (T-18 at 13:00):** Once satisfied, the Board member submits the final one-pager to the Head of the MPD for the official record. The Head of MPD does not approve or alter content. These documents are disclosed publicly with a 12-year lag under the CBA's transparency strategy.

The full set of one-pagers forms a key input into the **Taxonomy of Scenarios in the Monetary Policy Report**.

Scenarios Meeting (T-15)

Roughly two weeks before Decision Day, the Board meets with the Policy Round Coordinator to review and approve two elements:

- The **Taxonomy of Scenarios**, summarizing all risks and narratives submitted by Board members.
- The proposed illustrative **Case A and Case B** scenario narratives to be developed and quantified.

The Board's role is to provide direction and ensure the taxonomy captures their concerns, not to specify technical assumptions. By the end of the meeting, Board members should have a clear mental picture of how scenarios will be built and how they will feed into the Monetary Policy Report.

Scenario-Building Phase

Following the Scenarios Meeting, the Policy Round Coordinator leads daily production meetings with staff. Together, they build out:

- the **market reference scenario**,
- the **illustrative Case A scenario**, and
- the **illustrative Case B scenario**.

These are quantified using the semi-structural quarterly projection model and satellite models where appropriate. Particular emphasis is placed on the policy implications — specifically, the interest-rate paths required to ensure convergence toward the inflation target.

The illustrative scenarios are not prescriptive. They do not constrain Board members' views or votes. Rather, they provide clear reference points for evaluating risks, identifying potential dark-corner outcomes, and structuring the narrative of the Monetary Policy Report.

Submission of Final Scenarios (T-3)

Three days before Decision Day, the Policy Round Coordinator submits the final Case A and Case B scenario narratives and their quantitative projections to the Board, along with the completed draft of the Monetary Policy Report. No further changes are made after submission.

Decision Day (T)

With the scenarios finalized and 28 days of discussion behind them, the Board convenes to make the policy decision. Staff present current and underlying economic conditions, along with the scenario implications. Board members then deliberate and vote.

Each Board member submits a brief explanation outlining:

- the policy action they support,
- the risks and assumptions driving their view, and
- their willingness to change course if new information emerges.

The decision is announced together with publication of the Monetary Policy Report, which explains the policy stance through a structured narrative grounded in the scenario framework. The Governor then holds a press conference, broadcast live to the public.

2. Avoiding the "Folly in Baselines"

Traditional FPAS Mark I frameworks relied heavily on deterministic baselines. However, this approach risked institutionalizing error by encouraging overconfidence in a single projection. FPAS Mark II warns against this "folly in baselines," observing that a central bank must operate under multiple plausible futures.

FPAS Mark II embeds uncertainty directly into analysis by requiring multiple scenarios. Each scenario represents a different possible reality shaped by alternative assumptions about shocks, transmission channels, and expectations. Policy must be evaluated not by whether the baseline materializes, but by how prudently the central bank manages the distribution of risks around it.

This captures the core of the least-regrets mindset: the objective is not precision but prudent risk management.

3. Policy-Relevant Analysis: Bridging Models and Judgment

Under FPAS Mark II, staff focus on producing policy-relevant analysis instead of perfect forecasts. Policy-relevant analysis is defined as translating analytical results into information that helps the Board understand risks and make decisions under uncertainty.

Good analysis:

1. Integrates model outputs and expert judgment into coherent narratives.
2. Quantifies uncertainty via scenario matrices.
3. Provides communication materials that clarify risks, trade-offs, and credibility issues.

Analysis is designed not to predict what the Bank expects, but to reveal what it fears and what it hopes for.

4. The Role of Staff–Board Interaction

The 29-day cycle intensifies collaboration between staff and Board. Staff no longer defend a single baseline; instead, they facilitate structured analytical dialogue. The Board becomes decision-makers under uncertainty, applying least-regrets reasoning to evaluate risks.

As the manual states: "Staff exist not to convince the Board, but to prepare the Board for disciplined disagreement." Constructive tension is expected and encouraged, as it sharpens the collective understanding of risks.

The Executive Briefing at the end of the cycle embodies this partnership - it is not a forecast but a framework to support decision-making under uncertainty.

5. Continuous Learning and Institutional Credibility

FPAS Mark II transforms the central bank into a learning institution. Each cycle ends with a structured review of forecast errors, communication performance, and structural shifts.

These lessons shape the next round. The central bank continually updates its models, judgment, and communication practices.

Credibility emerges not from avoiding mistakes but from demonstrating the capacity to learn from them quickly. Human capital development is embedded in the policy process itself.

6. Policy Implications and Concluding Reflections

The transition to the 29-day cycle and policy-relevant analysis has several implications:

- Predictability and Rhythm: The cycle creates discipline and enhances credibility.
- Transparency and Accountability: Transparency about risks and how the central bank manages them form credibility.
- Resilience: Continuous learning enables flexible responses to shocks.
- Human Capital: Staff development is part of the policy system, not an external function.
- Prudent Risk Management: Decisions weigh risks across multiple futures.

As summarized in FPAS Mark II: "The future cannot be forecast, but it can be managed. The discipline of the policy cycle and the humility of least-regrets thinking make credibility sustainable."

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Script

Hello everyone, I'm **Susanna Grigoryan** from the **Central Bank of Armenia and a Level One student at the Global Forecasting School**. Welcome back to the CBA Academy Learning Hub. In this session, we will walk through how the 29-day policy cycle works under FPAS Mark II, and how it transforms analysis, judgment, and institutional learning.

FPAS Mark II starts from the recognition that uncertainty is the norm, not the exception. Forecasting is no longer about predicting a single path for inflation or output. It is a disciplined process of learning under uncertainty. This philosophy shapes the structure of the 29-day cycle, where analysis, scenario building, judgment, and communication are tightly integrated.

The cycle begins with a concise staff briefing on day T–29. This Kick-Off Meeting presents the main economic drivers, recent developments, and the key uncertainties shaping the environment. The briefing is followed by an open discussion among Board members about the major risks they see ahead. The purpose is not to dwell on the latest data release but to surface the deeper concerns, tail risks, and vulnerabilities that will frame the entire round.

Over the next two weeks, each Board member develops a one-page narrative—an Initial Macroeconomic Risk Assessment—outlining the risks, uncertainties, and issues they believe must be managed in the current round. These one-pagers reflect the 3R principle: scenarios must be related to current conditions, realistic enough to merit consideration, and relevant for monetary policy. They are not binding documents. Board members are encouraged to revise their thinking as the round unfolds. This flexibility is a deliberate feature of FPAS Mark II.

To support this process, each Board member works with two staff coordinators. They hold an initial brainstorming session where Board members outline their preliminary thoughts. Coordinators help structure these ideas, ensure they align with the Three-R criteria, and connect them to insights from the Kick-Off Meeting and weekly analytical briefings. The coordinators then prepare a first draft of the one-pager, and over the following days the draft is refined through short iterations. By T–18, the final version is submitted to the Head of the Monetary Policy Department for the official record. These documents later feed directly into the taxonomy of scenarios.

Roughly two weeks before the policy decision, the Board meets for the Scenarios Meeting. The Policy Round Coordinator presents the consolidated taxonomy of scenarios, which reflects all the risk assessments submitted by Board members. The coordinator also presents the proposed narratives for the illustrative Case A and Case B scenarios that will be quantified for the round. Board members are not responsible for technical assumptions. Their task is to confirm that the taxonomy captures the risks that matter most to them. By the end of the meeting, everyone has a clear understanding of how the scenarios will be developed and how they will shape the Monetary Policy Report.

From that point on, the scenario-building phase begins in earnest. The Policy Round Coordinator leads daily production meetings with staff. Together they construct the market-reference scenario, the Case A scenario, and the Case B scenario. These are quantified using the quarterly projection model and satellite models where appropriate. The focus is on the policy implications of each scenario, especially the interest-rate paths needed to return inflation to target. These illustrative scenarios are not intended to constrain Board members. They are reference points that help evaluate risks, identify potential dark-corner outcomes, and structure the narrative that will later appear in the Monetary Policy Report.

Three days before the policy decision, the Policy Round Coordinator submits the final Case A and Case B narratives, along with their quantitative projections, to the Board together with the full draft of the Monetary Policy Report. No further changes are made after this point.

Decision Day brings together four weeks of analysis, debate, and scenario work. Staff present the current and underlying economic conditions and explain how the different scenarios map onto the risks facing the economy. Board members then deliberate on the appropriate policy stance. Each Board member submits a concise explanation of their vote, detailing the action they support, the risks and assumptions that shape their view, and their willingness to adjust their position if new information emerges. The decision is announced together with the publication of the Monetary Policy Report, which explains the policy stance through a clear narrative grounded in the scenario framework. The Governor then holds a press conference, live-streamed to the public.

FPAS Mark II deliberately avoids the old “folly in baselines.” Earlier frameworks relied heavily on a deterministic baseline forecast, which encouraged excessive confidence in a single view of the world. FPAS Mark II replaces that mindset with structured scenario analysis. Each scenario represents a different plausible reality shaped by alternative shocks, transmission channels, and expectations. The aim is not precision, but prudent risk management. What matters is not whether the baseline materializes, but how effectively the central bank manages the distribution of risks across possible futures.

Under this framework, staff focus on policy-relevant analysis rather than on perfect projections. Policy-relevant analysis integrates model outputs and economic judgment into coherent narratives, quantifies uncertainty, and presents the risks and trade-offs in a way that helps the Board deliberate under uncertainty. It is not about predicting what the Bank expects, but about illuminating what it fears and what it hopes to avoid.

The 29-day cycle also deepens collaboration between staff and the Board. Staff no longer defend a single baseline; they facilitate structured analytical dialogue. The Board becomes a decision-making body that weighs risks using the least-regrets principle. Constructive tension is expected and even desirable, because it sharpens the institution's understanding of risks. The Executive Briefing at the end of the cycle captures this partnership: it is not a forecast but a framework to support disciplined decision-making.

FPAS Mark II turns the central bank into a learning institution. Each cycle ends with a structured review of forecast performance, communication outcomes, and shifts in economic relationships. These lessons feed directly into the next round. Credibility grows not because the central bank avoids mistakes, but because it learns from them quickly and openly. Human capital development is not an external goal — it is built into the cycle itself.

The move to the 29-day cycle and policy-relevant analysis has important implications. It creates a predictable rhythm that strengthens credibility. It separates analysis from judgment, improving transparency and accountability. It supports resilience by allowing the institution to adjust quickly to shocks. And it embeds prudent risk management in every stage of policymaking.

FPAS Mark II reminds us that the future cannot be forecast, but it can be managed. The discipline of the policy cycle and the humility of least-regrets thinking make credibility sustainable.

Thank you for watching, and I would like to acknowledge Douglas Laxton, Jared Laxton, Asya Kostanyan, and Sopio Mkervalidze at the Global Forecasting School for their guidance and support in producing this video.

See you there.

Video 13 - Analytical Toolkit and Modelling Innovations

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November 2025

Note

1. Limitations of FPAS Mark I

FPAS Mark I relied on a deterministic baseline and a single linear core model. While useful for transparency, it could not capture nonlinearities, asymmetric risks, credibility dynamics, or rapid shifts in financial conditions. These limitations motivated the development of FPAS Mark II, which is designed to better support policy under uncertainty.

Since the onset of the COVID-19 pandemic, the global policy environment has been characterized by a level of uncertainty that has become—using Alan Greenspan's words—the "defining feature of the monetary policy landscape." A series of overlapping shocks, including geopolitical tensions, energy-price volatility, and supply-chain disruptions, has reinforced this uncertainty and exposed the limits of relying on a single baseline projection. The world economy has become more vulnerable to nonlinear dynamics and high-impact tail events, prompting a fundamental reassessment of how central banks forecast and analyze the economy.

These developments were a major catalyst for the design of FPAS Mark II and also highlighted the need to upgrade the analytical tools that support monetary-policy analysis. A modern framework must be equipped to construct scenarios, test sensitivities, and examine the credibility implications of alternative policy paths.

The earlier generation of FPAS, developed by Berg, Karam, and Laxton (2006), transformed monetary-policy analysis across central banks. Its core tool, the Quarterly

Projection Model (QPM), provided a structured, accessible way to support inflation-targeting regimes. QPMs rely on New Keynesian principles—nominal and real rigidities, business-cycle behavior, and rational expectations—combined with a policy rule that guides interest-rate adjustments to stabilize inflation and anchor expectations. The CBA adopted this approach in 2006, in line with many central banks transitioning to inflation targeting.

However, these conventional frameworks have important limitations. They rely on linear relationships that may break down during periods of economic stress, and they treat credibility as fixed and external to the model. In reality, credibility evolves with policy performance and can deteriorate quickly. Ignoring these nonlinearities creates risks. For example, when the output gap becomes strongly positive, the Phillips Curve may steepen, causing inflation expectations to accelerate more rapidly than a linear model would suggest. Failing to recognize this can lead to delayed policy action, requiring more aggressive tightening later and resulting in larger economic costs. The same logic applies to credibility: it can erode suddenly but is difficult and costly to rebuild.

In a world of heightened uncertainty and competing interpretations of the data, an analytical framework that captures these nonlinearities becomes essential. It allows policymakers to better understand the full consequences of potential policy missteps and to make decisions that minimize regrets. For this reason, nonlinear analytical structures—and the risk-management perspective they support—are a core component of the Prudent Risk Management Approach to Price Stability.

2. Nonlinearities and State Dependence

FPAS Mark II incorporates nonlinear relationships-particularly in the Phillips Curve, the transmission mechanism, and credibility dynamics. Inflation responds more strongly in overheating conditions than in slack periods, implying that late tightening is more costly than early measured action. Nonlinearities help policymakers model real-world behaviour more accurately and capture asymmetric risks inherent in modern monetary policy.

3. Endogenous Credibility

FPAS Mark II models credibility as an evolving stock influenced by central banks ability to meet its targets,its communication quality, and institutional consistency. Expectations adjust as the central bank demonstrates competence and reliability. Modelling credibility endogenously strengthens the connection between current policy actions and future outcomes, recognizing that credibility must be earned continuously.

4. The FPAS Mark II Model Suite

FPAS Mark II employs a suite of complementary models rather than relying on a single structure. These include:

- **EndoCred Model:** Incorporates endogenous credibility, nonlinear Phillips Curve dynamics, and asymmetric loss functions.
- **MPMOD:** Informs estimates of latent variables, such as the output gap, with theoretical relationships linking unobservable with observable variables.
- **FCMOD:** Captures financial cycle dynamics and systemic vulnerabilities.
- **Linear-QPM:** Retained for continuity, historical decompositions, and pedagogical comparison.
- **Monthly Projection Model:** Used for thinking of economic scenarios using data that is available on more frequent basis.
- **Dynamic Stochastic General Equilibrium Models (DSGE):** Examples are GIMF-type models used for medium-term policy analysis, especially when fiscal and monetary interactions matter.
- **GPM:** 3-country (US, EU, Russia) plus commodities (oil, copper, food) linear model used for analyzing the global sector

Using a model suite helps diversify analytical risks and provides multidimensional insight into inflation, output, and financial conditions.

ENDOCRED Model

The ENDOCRED model builds on the foundations of the traditional QPM introduced by Berg, Karam, and Laxton (2006), while introducing several critical innovations suited to a world of heightened and persistent uncertainty. A central feature of the model is the endogenization of policy credibility. Instead of treating credibility as fixed, ENDOCRED allows it to evolve over time in a nonlinear way, directly influenced by the central bank’s past performance. This dynamic structure captures the reality that credibility can erode quickly when policy missteps occur and strengthens only gradually when policy is consistent and disciplined.

Inflation dynamics are modeled through a convex Phillips curve, which responds more strongly when inflation is high or when the output gap becomes significantly positive. This specification reflects real-world behavior more accurately than a linear formulation, particularly during periods of overheating or when expectations begin to de-anchor.

The IS curve is also extended beyond its traditional short-term focus. ENDOCRED introduces a long-term effective interest rate derived from yields across different maturities based on the expectations hypothesis and a time-varying term premium. This feature allows the model to incorporate the broader and more persistent influence of monetary policy on economic activity, capturing the role of the yield curve in shaping demand conditions.

In contrast to the standard Taylor-type policy rule used in many QPMs, ENDOCRED adopts a loss-function-based policy design. This enables more flexible, state-contingent responses, which are essential when uncertainty is elevated, transmission is impaired, or the economy is operating near policy limits.

Another key innovation is the explicit treatment of dollarization. The influence of external interest rates on the domestic effective interest rate depends on a dollarization parameter that is inversely related to policy credibility. As credibility improves, dollarization declines, strengthening the effectiveness of domestic monetary policy. This creates a meaningful feedback loop: stronger credibility reduces dollarization, which enhances transmission, which in turn helps credibility rise further.

Together, these innovations allow ENDOCRED to capture nonlinearities, credibility dynamics, and the broader channels through which monetary policy affects the economy—making it an analytical framework well suited for the Prudent Risk Management Approach.

QPM Model

The Quarterly Projection Model (QPM) has long been the traditional workhorse of FPAS Mark I institutions operating under inflation targeting. While ENDOCRED now serves as the primary policy model at the CBA, the QPM continues to play an important complementary role within the analytical toolkit. As a linear semi-structural model, QPM is particularly useful for filtering historical data and for generating internally consistent estimates of key unobservable variables. These include the neutral interest rate, the real effective exchange rate gap, and the output gap—core indicators needed to assess the cyclical position of the economy and to anchor policy discussions.

GPM Model

The Global Projection Model (GPM) also remains a valuable part of the CBA's modeling architecture, particularly for analyzing Armenia's external environment. GPM is a three-country model that captures the economies most relevant for Armenia: the United States, the euro area, and Russia. The model generates coherent projections for these major partners, allowing staff to assess external demand, global inflationary pressures, and interest-rate dynamics that shape Armenia's open-economy conditions.

In addition to its country blocks, GPM incorporates commodity prices that are especially important for Armenia's macroeconomic outlook. These include copper—one of Armenia's significant export products—as well as food and oil, both of which play sizeable roles in domestic inflation, given Armenia's reliance on imported food and its status as an oil importer. By providing structured projections for these external drivers, GPM offers a clear frame of reference for understanding external risks and evaluating how global developments propagate into the domestic economy under the FPAS Mark II scenario-based framework.

5. Analytical Dashboards

High-quality current economic analysis is the foundation of the Monetary Policy Department's work. It is conducted continuously and presented to the Board during weekly analytical briefings every Wednesday, which cover all policy-relevant developments in both the global and domestic economy. To strengthen efficiency, consistency, and analytical rigor, the Monetary Policy department has developed a series of technical tools—foremost among them, the analytical dashboards.

These dashboards cover a wide range of economic themes: the labor market, inflation dynamics, external developments, sectoral growth patterns, and the decomposition of macroeconomic projections into production and expenditure components, among others. Each dashboard serves as a centralized platform that automatically draws data from the department's unified data warehouse, transforms the raw series into analytically meaningful variables, and presents them in a format designed for fast and rigorous assessment.

The dashboards generate several important benefits. First, they dramatically **increase efficiency**. By automating data extraction and calculation, they eliminate much of the manual work associated with gathering frequently used indicators. This allows staff to devote more time to deeper analysis—whether that involves understanding the drivers of inflation, assessing the main contributors to growth, or conducting ad-hoc investigations during periods of heightened uncertainty.

Second, dashboards greatly facilitate **knowledge transfer** between rotating teams. One of the persistent challenges of a quarterly policy cycle is ensuring continuity when staff teams change. Dashboards provide a unified analytical structure built from common data and standard transformations. New teams can continue the work of their predecessors without having to rebuild datasets or replicate previous calculations, ensuring continuity in both analysis and narrative development.

Finally, dashboards **simplify several complex, recurring tasks** that previously required substantial time and effort. A clear example is the decomposition of GDP growth into expenditure components—an essential step in translating output from the ENDOCRED model into detailed macroeconomic stories. What was once a labor-intensive process is now automated through a dedicated dashboard, saving considerable staff time and reducing the risk of inconsistencies. Similar efficiencies have been achieved across other standardized processes.

Together, these analytical dashboards create a more efficient, consistent, and resilient analytical environment.

6. Underlying Inflation Measures (NTSPI)

While headline CPI remains the primary benchmark for monetary policy, its individual components provide critical insight for risk-management-based policymaking. Effective policy discussions—and credible communication—require understanding which components of inflation matter most for transmission, for expectations, and for identifying emerging risks. This raises an important analytical question: *how should the CPI be decomposed to help policymakers make better decisions?*

This question was central to the research on **Non-Traded Sticky Price Inflation (NTSPI)**, which designs a CPI decomposition grounded in modern monetary policy transmission. The aim is to separate components driven mainly by supply and exchange-rate movements from those driven by domestic demand and inflation expectations—the latter being essential for evaluating medium-term price stability.

Traditional underlying inflation measures, such as core, median, or trimmed-mean inflation, rely mostly on statistical filters. Their main purpose is to remove volatility, not necessarily to distinguish items based on economic transmission channels. In contrast, New Open Economy Macroeconomics provides a conceptual rationale for classifying CPI components into **flexible** and **sticky** categories.

Flexible prices are those that respond quickly to shocks—especially exchange-rate movements and supply-side disturbances. These are typically internationally traded goods. Sticky prices, by contrast, adjust more slowly. They often reflect domestic services and non-traded goods, where price-setting is less frequent and more influenced by demand conditions and expectations.

This distinction traces back to the classic Dornbusch overshooting model, where asset prices move quickly while most goods prices adjust sluggishly. Traded goods fall into the flexible bucket because they respond almost immediately to exchange-rate changes.

Non-traded goods—such as housing, health, education, and many services—fall into the sticky bucket precisely because their prices adjust more gradually.

Understanding the behavior of these two buckets is essential. Sticky prices respond more slowly to economic conditions, making them informative about the persistence of inflation and the anchoring of expectations. Flexible prices, on the other hand, move quickly in response to idiosyncratic or external shocks. While these shocks may be temporary, they can signal emerging inflationary pressures—particularly when many flexible components start rising simultaneously due to generalized excess demand.

If monetary policy fails to respond in time, these pressures may spill over into sticky components such as wages and services. Once sticky prices begin to accelerate, reversing the trend becomes much more costly, often requiring a significantly more aggressive tightening that imposes unnecessary harm on the real economy.

The underlying challenge is that CPI aggregates combine prices from diverse markets. Inflation becomes meaningful for policy when many markets collectively indicate excess demand—not when one item is volatile. A decomposition grounded in economic transmission helps policymakers distinguish between transient volatility and genuine underlying pressure.

In Armenia, the NTSPI basket includes non-traded goods and services such as housing, healthcare, education, utilities, transportation, and communication services. As of 2023, these items make up roughly **15 percent** of the Armenian CPI. Monitoring this basket provides valuable information about domestic inflation momentum, expectations, and the degree to which external shocks may be feeding into more persistent components.

7. Integrating Judgment and Model-Based Evidence

FPAS Mark II emphasizes structured judgment - expressed through scenario design, risk mapping, credibility assessments, and interpretation of model outputs. Judgment does not override models but complements them. Staff must explain how judgment modifies model projections, increasing transparency and strengthening policy briefings.

8. Implications for Small Open Economies

Armenia is highly exposed to external shocks such as commodity prices, global markets, and geopolitical developments. FPAS Mark II equips policymakers to analyse nonlinearities, asymmetries, and risk distributions more effectively critical tools for small open economies.

By reducing vulnerability to forecast errors and improving resilience to external shocks, FPAS Mark II strengthens macroeconomic stability.

Conclusion

The Analytical Toolkit of FPAS Mark II transforms forecasting and policymaking into a structured, nonlinear, credibility-aware learning system. By integrating models, dashboards, improved measures of underlying inflation, and structured judgment, FPAS Mark II enables least regrets decision-making under uncertainty and enhances institutional credibility.

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Script

Hello everyone, I'm Susanna Grigoryan from the Central Bank of Armenia and a Level One student at the Global Forecasting School. In this session, we'll look at how the analytical toolkit of FPAS Mark II has evolved, and how it helps us make better policy decisions under uncertainty.

Under FPAS Mark I, most central banks relied on a deterministic baseline and a single linear core model. That framework was extremely valuable for transparency and for building modern inflation-targeting regimes. At the CBA, as in many other central banks, the Quarterly Projection Model introduced by Berg, Karam, and Laxton became the main

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workhorse. It combined New Keynesian features like rigidities and rational expectations with a policy rule that adjusted interest rates to stabilize inflation and anchor expectations. This was a major step forward when inflation targeting was introduced in Armenia in 2006.

But the world has changed. Since the onset of the COVID-19 pandemic, the global policy environment has been shaped by an very high and persistent level of uncertainty. Alan Greenspan once called uncertainty the “defining feature of the monetary policy landscape,” and recent years have made that description even more accurate. We have seen overlapping shocks from pandemics, geopolitical tensions, energy price swings, and supply-chain disruptions. The global economy has become more susceptible to nonlinear dynamics and high-impact tail events, and this has exposed the limits of relying on a single linear baseline projection.

Traditional FPAS frameworks assume linear relationships and treat credibility as something fixed in the background. In reality, credibility is earned and lost over time. When the output gap is strongly positive, for example, the Phillips curve can steepen, and inflation expectations can accelerate faster than linear models would suggest. If policy reacts too late, the adjustment costs for inflation and output become much higher. The same logic applies to credibility: once it starts to erode, rebuilding it is slow and costly. In this kind of world, a framework that ignores nonlinearities and credibility dynamics underestimates the true risks of policy mistakes.

These developments were a major catalyst for FPAS Mark II. The new framework recognizes that uncertainty is structural and that risk management—not point prediction—must guide policy. That means we need tools that can help build realistic scenarios, test sensitivities, and assess how different policy paths affect credibility over time. Nonlinear structures and explicit risk management are no longer optional add-ons; they are central to how we think about policy.

FPAS Mark II therefore incorporates nonlinearities and state dependence directly into the analytical core. Inflation does not respond in the same way in every state of the world. In overheating conditions, when the output gap is large and positive and expectations are at risk of de-anchoring, inflation responds much more strongly to demand than it does in periods of slack. A nonlinear Phillips curve captures this reality and shows why late tightening can be far more costly than early and measured action. The same applies to financial conditions, transmission channels, and credibility: the impact of a small policy error is not constant across states of the world.

Credibility itself is treated as an endogenous variable. In FPAS Mark II, credibility is modeled as a stock that evolves over time with the central bank's actions. It depends on the ability to achieve the inflation target, on the consistency of decisions, on the quality of communication, and on institutional behavior more broadly. As the central bank demonstrates competence and reliability, expectations adjust and credibility strengthens. When policy is inconsistent or poorly explained, credibility weakens. Treating credibility this way tightens the link between today's decisions and tomorrow's outcomes and reinforces the idea that credibility must be earned continuously, not assumed.

To support this way of thinking, FPAS Mark II does not rely on a single model. It uses a suite of complementary models, each serving a specific purpose. The ENDOCRED model is at the center. It builds on the traditional QPM but endogenizes credibility, introduces a convex Phillips curve, and extends the IS curve to include a long-term effective interest rate derived from the yield curve and a time-varying term premium. It uses a loss-function-based policy design instead of a simple Taylor rule, and it explicitly links dollarization to credibility, so that as credibility improves, dollarization declines and the effectiveness of domestic monetary policy increases.

Alongside ENDOCRED, the CBA uses other models that play distinct roles. MPNOD helps estimate latent variables such as the output gap and neutral interest rate by linking unobservable concepts to observed data. A financial-cycle model-FCMOD-focuses on credit, leverage, and systemic vulnerabilities. The linear QPM is retained as a complementary tool for filtering history and extracting consistent estimates of unobservables, especially useful for historical decompositions and pedagogical comparison. A monthly projection model helps think about near-term scenarios using higher-frequency data. DSGE-type models, such as GIMF-style structures, are used for medium-term policy questions where fiscal and monetary interactions are central. And the Global Projection Model provides a structured representation of Armenia's main external partners—the United States, the euro area, and Russia—as well as key commodities like oil, copper, and food, all of which are crucial for a small open economy like Armenia.

Using a model suite spreads analytical risk. No model is treated as the "truth." Instead, each model provides a different lens, and enhances policy discussions.

The analytical infrastructure supporting these models is just as important. High-quality current analysis is carried out continuously and presented to the Board every Wednesday. To make this work more efficient and consistent, the Monetary Policy Department relies heavily on analytical dashboards. These dashboards draw automatically from a unified data warehouse and transform raw series into meaningful indicators across key areas—labor markets, inflation, external conditions, sectoral growth, and the breakdown of GDP into expenditure and production components.

By automating data collection and transformation, dashboards drastically reduce the time staff spend on repetitive work and free them to focus on interpretation and risk assessment. They also make team rotations much smoother. When staff move between teams, the dashboards ensure that data, methods, and key indicators are consistent over time, so new teams can pick up where previous teams left off without rebuilding everything from scratch. Complex recurring tasks, such as decomposing GDP growth coming from ENDOCRED into detailed demand-side stories, are now managed through dedicated dashboards instead of manual spreadsheets. Beyond efficiency, dashboards integrate high-frequency indicators, model outputs, financial conditions, sectoral data, risk distributions, and scenario trees. They help staff present risk-aware assessments and allow the Board to visualize uncertainty in a structured and intuitive way.

Another important improvement in the FPAS Mark II toolkit is the treatment of underlying inflation. Headline CPI is still the main policy benchmark, but its components carry crucial information for risk management. In Armenia, the Non-Traded Sticky Price Inflation index—NTSPI—has been developed to distinguish between flexible and sticky components in a way that reflects the transmission mechanism. Flexible prices, typically found in traded goods, adjust quickly to exchange-rate and supply shocks. Sticky prices, often in non-traded services like housing, healthcare, and education, adjust slowly and are more closely tied to domestic demand and expectations. Monitoring NTSPI, which covers non-traded goods and services such as utilities, transportation, communication, and other services that make up around 15 percent of the CPI basket, gives policymakers a clearer sense of persistent inflation pressures and whether temporary shocks are spilling over into components that are harder to reverse.

All of this feeds into a broader principle: FPAS Mark II is not about replacing judgment with models; it is about structuring judgment. Judgment appears in the design of scenarios, in the mapping of risks, in the interpretation of model outputs, and in the assessment of credibility. Staff are expected to explain how and why their judgment modifies model projections, making the reasoning behind recommendations transparent. The Board, in turn, uses this structured analysis to apply least-regrets thinking when it votes on policy.

For a small open economy like Armenia, exposed to external shocks, commodity price swings, and geopolitical developments, this approach is especially important. FPAS Mark II equips policymakers to analyze nonlinearities, asymmetries, and risk distributions more effectively. It reduces vulnerability to forecast errors and strengthens the institution's ability to navigate external shocks without over-reacting or under-reacting.

In summary, the analytical toolkit of FPAS Mark II turns forecasting and policymaking into a structured, nonlinear, credibility-aware learning system. By combining model suites, dashboards, improved measures of underlying inflation, and structured judgment, FPAS Mark II supports least-regrets decision-making under uncertainty and helps sustain institutional credibility over time.

Thank you for watching, and I would like to acknowledge Douglas Laxton, Jared Laxton, Asya Kostanyan, and Sopio Mkervalidze at the Global Forecasting School for their guidance and support in producing this video.

See you there.

Video 14 - Dynamic Learning and Human Capital Development

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November 2025

Note

1. Human Capital as a Core Component of FPAS Mark II

FPAS Mark II transforms not only analytical tools but also the institutional processes through which policymakers learn, collaborate, and develop expertise. For a small open economy like Armenia-where uncertainty is persistent and shocks frequently originate from external sources-human capital becomes a core pillar of the policy framework.

The FPAS Mark II manual emphasizes that credibility is sustained not only through correct decisions, but through the institution's demonstrated ability to learn continuously. Staff must be capable of integrating modelling results, scenario analysis, financial data, and judgment, a skillset cultivated through structured training and experience.

Human capital is treated as a macroeconomic asset: the quality of staff judgment determines the quality of policy decisions.

2. Continuous Learning and Institutional Memory

The FPAS Mark II framework embeds learning directly into the 29-day policy cycle. After each round, teams conduct structured post-mortems evaluating:

- forecast accuracy,
- scenario relevance,
- communication clarity,
- interpretation of risks, and
- the behaviour of key relationships (e.g., Phillips Curve, expectations, transmission channels).

The emphasis is not on avoiding mistakes but on learning faster than others. Each cycle creates institutional memory, ensuring that lessons carry into the next round. This enhances credibility by demonstrating adaptability and humility-key virtues in an uncertain world.

3. The Rotation System: Building Versatility and Reducing Silos

FPAS Mark II introduces a structured rotation mechanism across teams. Analysts rotate through forecasting, scenario development, external-sector monitoring, financial-cycle analysis, and inflation decomposition. Rotations achieve several goals:

- prevent analytical silos,
- broaden staff expertise,
- strengthen institutional resilience, and
- ensure continuity even when key individuals change roles.

By broadening skillsets and promoting cross-team understanding, the rotation system supports a cohesive institutional culture centred on learning.

4. Internal Training Routines and Capacity Building

Learning is reinforced by internal routines designed to ensure continuous development:

- Daily analytical briefings
- Weekly seminars on FPAS tools and modelling
- Monthly cross-team workshops
- Case-based exercises using past policy rounds
- Internal courses on expectations, nonlinearities, and risk assessment.

These routines create a pipeline of future leaders capable of presenting scenarios, preparing executive briefings, and guiding the Board through uncertainty.

5. International Collaboration and External Learning Channels

FPAS Mark II places strong emphasis on global learning. Staff participate in:

- regional workshops,
- collaborations with partner central banks,
- international seminars and conferences, and
- research exchanges with academic and policy institutions.
- External exposure broadens perspectives, introduces new analytical tools, and keeps the institution aligned with global best practices in uncertainty-aware policymaking.

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6. Learning as a Credibility Mechanism

Credibility in FPAS Mark II is driven not only by outcomes but by demonstrated learning. Clear acknowledgment of past errors and transparent communication of improvements strengthen public trust.

A central bank that learns continuously:

- upgrades its analytical capacity,
- upgrades its analytical capacity,
- reduces vulnerabilities, and
- builds confidence in its long-run policy framework.

Human capital is not a complement to FPAS Mark II-it is one of its foundations.

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Script

Hello everyone, I'm **Susanna Grigoryan**, an economist at the **Central Bank of Armenia** and a **Level One student at the Global Forecasting School**. In this session, we explore how FPAS Mark II transforms the Central Bank of Armenia into a modern learning institution—one where human capital, knowledge sharing, and continuous improvement stand at the centre of credible monetary policy.

FPAS Mark II transforms not only the analytical framework of monetary policy but also the institutional processes through which policymakers learn, collaborate, and develop expertise. For a small open economy like Armenia, where uncertainty is persistent and shocks originate from multiple external sources, human capital becomes a core pillar of the monetary policy framework. The FPAS Mark II manual emphasizes that disciplined learning is central to sustaining credibility, strengthening judgment, and improving the quality of policy decisions over time.

Under FPAS Mark II, analytical tools and models are only as effective as the people who use them. The manual stresses that credibility comes not only from correct decisions but from the institution's demonstrated ability to learn and improve. This requires staff who can integrate modelling results, scenario analysis, financial data, and judgment-based narratives—a skill set developed over time through structured learning. FPAS Mark II positions human capital as a macroeconomic asset. Staff skills determine whether the institution can correctly diagnose risks, interpret uncertainty, and support least-regrets decisions.

The Mark II framework embeds learning directly into the policy cycle. After each round, teams conduct post-mortems assessing forecast accuracy, scenario usefulness, communication effectiveness, and risk interpretation. This ensures that lessons are carried into the next cycle, creating an institutional memory. The manual underscores that central banks cannot avoid mistakes—but they can demonstrate an ability to learn faster than others. This learning capacity is central to long-term credibility and to sustaining anchored expectations in a volatile world.

FPAS Mark II also introduces a structured rotation mechanism across teams. Analysts rotate between forecasting, scenario development, external conditions monitoring, and specialized modules such as the financial cycle or inflation decomposition. This increases resilience by ensuring that institutional knowledge does not rest with a few individuals. Rotations prevent analytical silos, broaden skill sets, and ensure that staff gain exposure to multiple components of the FPAS system.

On-the-job learning is reinforced by structured training routines that include daily analytical briefings, weekly seminars on FPAS tools, monthly cross-team workshops, case-based exercises using past policy rounds, and internally taught courses on modelling, expectations, and risk assessment. These routines ensure that learning is continuous rather than sporadic. They also prepare staff to take on roles with higher responsibility, including presenting scenarios, leading policy workshops, and drafting executive briefings.

The Mark II framework also places strong emphasis on global learning. Staff participate in regional workshops, collaborate with partner institutions, present analytical work in international seminars, and engage with research communities. The manual views external learning as essential—exposing staff to new methodologies, alternative frameworks, and global best practices in uncertainty-aware policymaking.

Credibility in FPAS Mark II is shaped not only by decisions but by the institution's demonstrated ability to adapt and improve. Transparent communication of learning, including acknowledgment of past errors, reinforces public trust. A central bank that learns is one that consistently upgrades its analytical capacity, reduces vulnerability to external shocks, and strengthens its long-term policy framework.

Dynamic learning and human capital development form the backbone of FPAS Mark II. By embedding learning routines, rotations, and continuous feedback loops into the policy process, the Central Bank of Armenia ensures that its staff remain adaptive, analytical, and capable of supporting risk-based decision-making. Human capital is not a complement to FPAS Mark II—it is one of its foundations.

Thank you for watching, and I would like to acknowledge **Douglas Laxton, Martin Galstyan, Jared Laxton, Asya Kostanyan, and Sopio Mkervalidze** at the **Global Forecasting School** and **The Better Policy Project** for their guidance and support in producing this video.

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! Video 15 - Organization, Rotation, and Performance Management

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Note

Under FPAS Mark II, good monetary policy is not only about models and scenarios— it is also about how people are organized, how they rotate across roles, and how their performance is managed and developed over time. A well-designed organizational structure ensures that the central bank's scarce human capital is used efficiently to support the least-regrets, scenario-based policy framework.

This session explains how the roles of the Chief Economist, the Director of Operations, Group Managers, projection round teams, and Board Coordinators fit together. It also shows how a structured rotation system across teams and roles helps sustain the FPAS processes, supports learning-by-doing, and strengthens motivation, collaboration, and productivity.

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Script

Hello everyone, I'm **Susanna Grigoryan**, an economist at the **Central Bank of Armenia and a Level One student at the Global Forecasting School.**

Today I'd like to walk you through how a modern Monetary Policy Department is organized under the FPAS Mark II framework, and how different roles work together to support high-quality forecasting, analysis, and decision making.

At the top of this structure is the Chief Economist, who also serves as the Head of the Monetary Policy Department. The Chief Economist is responsible for making sure all of the central bank's analytical resources—people, models, data, and time—are used efficiently to support the FPAS Mark II process. This person leads the main projects and processes in the department, and on a rotating basis coordinates and leads one of the quarterly projection rounds. The Chief Economist is also a key bridge between the Board and the staff, giving direction to the rest of the leadership team, including the Director of Operations and group managers, on both analytical and human-capital matters. To succeed in this role, the Chief Economist needs not only unquestionable macroeconomic expertise and a deep understanding of monetary policy, but also the ability to manage, lead, and inspire people every day.

Supporting the Chief Economist is the Deputy Head of the Monetary Policy Department, also referred to as the Director of Operations. This is a highly skilled macroeconomist who focuses on the operational side of the department. The Director of Operations manages the department's core assets—its models, documentation, data, and procedures—oversees how staff are allocated across tasks, and helps shape training, testing, and feedback processes. This role also involves building platforms for knowledge sharing, such as brown-bag lunches, seminars, workshops, and conferences, and coordinating closely with other parts of the central bank, from IT, cybersecurity, and HR to statistics, research, and financial stability. In many ways, the Director of Operations is the best of the group managers: someone who is technically strong but also excels at organizing work and supporting others so that the whole system runs smoothly.

Working with them are the administrative group managers. These are advanced macroeconomists who have mastered the core competencies of FPAS Mark II and reached higher certification levels. But what truly defines their role is not just technical ability; it's their skill in managing people and developing human capital. We think of managers not as "bosses" in a narrow sense, but as leaders who coach staff, help them grow, and create a supportive team culture. Group managers still do analytical work themselves, but a significant share of their time goes into training and coaching staff during the FPAS process, collaborating with them on products, helping to manage resources across teams, and providing ongoing feedback, including performance reviews and updates of job responsibilities. To be effective, they need to be inspirational, people-oriented, and committed to building one-on-one relationships, fostering teamwork and collegiality, and helping others succeed. This is not about being the smartest economist in the room; it's about nurturing a strong, healthy, high-performing team.

At the core of the department are the macroeconomists. They are the engine of FPAS Mark II and form the foundation of the team's analytical capacity. Macroeconomists are responsible for building and running models, drafting monetary policy reports, conducting cutting-edge research on key macroeconomic topics, and performing current analysis and scenario building. To operate at FPAS Mark II standards, they must be on track to becoming among the best in the world at what they do. This means having strong skills in current analysis, scenario design and policy analysis, clear communication, and modern macroeconomic theory and practice. They are expected to be fluent in both Armenian and English, have excellent command of advanced mathematics, and at least a working knowledge of coding languages such as R, Python, MATLAB, or Julia. Underlying all of this is the most important trait: critical thinking. FPAS Mark II macroeconomists cannot simply accept that "the model says X, therefore it must be true," or that a finding is correct just because famous economists have written about it. They must develop and defend their own views through rigorous reasoning and empirical analysis.

In addition to the internal staff, the framework also makes space for discrete project-based contract hires. Not all highly skilled macroeconomists, statisticians, or modelers can or want to commit to a full-time career path within the MPD, which requires being a team player and investing heavily in communication skills. However, their specific expertise can still be extremely valuable. To capture this, the department can bring such experts in for fixed-term projects, especially when highly specialized skills are needed that the internal team does not yet fully possess, or when time and resources are constrained. These contract hires might come from other central banks, international financial institutions, universities, or research institutions abroad. They can help with model development, specialized analysis, or programming, and an equally important contribution is the training and research collaboration they provide to the FPAS team.

A crucial part of this organizational structure is how staff are arranged during projection rounds. The Monetary Policy Department is organized into four teams, each made up of six macroeconomists. During each projection round, two of these teams are actively involved. One team takes the lead and does the "heavy lifting" by building and running the main scenarios—Case A, Case B, and, when necessary, additional cases like Case X or Y. They also draft the sections of the Monetary Policy Report that correspond to these scenarios and assist in preparing the Board presentation. The second team plays a supporting role, providing tactical and technical support, and helping the lead team where needed. Each round is led by a Projection Round Team Lead, chosen on a rotational basis, who serves as the thought leader for that cycle and is responsible for guiding the team through the daily work of the projection round and presenting content to the Board. This role is different from that of the group managers: the Team Lead's focus is on leading the analysis and narrative of the round, while group managers focus on people and resource management. Throughout the round, the Team Lead also receives support and guidance from the Chief Economist.

Within the leading six-person team, responsibilities are structured carefully. Four macroeconomists are assigned to work on the case scenarios and take full ownership of the scenario-building process, as well as the relevant sections of the Monetary Policy Report. The remaining two members of the team serve as coordinators for individual Board members. They work directly with Board members from the kick-off meeting through to decision day, helping to translate the Board's concerns, ideas, and views into clear, structured one-page narratives that capture their understanding of the economy, the main drivers, and possible policy responses. These one-pagers are drafted in both the native language and English where relevant and then submitted to the Chief Economist and the rest of the projection team, ensuring that the Board's perspectives are properly integrated into the scenarios and policy discussions.

After the projection round is completed, the case scenarios are locked in, the policy decision is taken, and the Monetary Policy Report is published. At that point, the round is formally over, and the focus shifts to preparing for the next policy cycle. The team that served as the back-up in the previous round now becomes the lead team for the upcoming projection. During the intervening period they take the lead in weekly macroeconomic monitoring meetings, prepare key analysis and stories that will feed into the next kick-off meeting, and continue work on research and improvements to the analytical toolkit. Over the course of a year, each of the four teams will lead a projection round once, serve as back-up once, and spend two quarters primarily on forward-looking work such as research, current analysis, and toolkit development. This rotation helps sustain the process over time, enhances productivity and efficiency, and supports a culture of collaboration, learning, and motivation across the department.

This is the essence of how FPAS Mark II envisions a modern Monetary Policy Department: with clearly defined, complementary roles; strong leadership; deep technical capacity; and a culture built on learning, critical thinking, and teamwork. Thank you for watching, and I hope this gives you a clearer picture of how all of these roles come together to support effective monetary policy.

Video 16 - Strategic Communication in Modern Central Banking

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November 2025

Note

Introduction

Strategic communication has become a central instrument of modern monetary policy. Under FPAS Mark II, communication is not supplementary-it is a core policy tool. Clear, transparent, and timely communication strengthens credibility, aligns expectations, and helps the public and financial markets interpret the central bank's risk assessments.

This chapter explores how FPAS Mark II reframes communication as an integral component of uncertainty management, expectations formation, institutional accountability, and monetary policy transmission.

1. Communication as a Policy Instrument

Earlier monetary policy regimes treated communication as an afterthought-mostly descriptive, not analytical. Under FPAS Mark II, communication becomes a true policy instrument. It must:

- shape expectations,
- clarify risks, and
- explain judgment under uncertainty.

Policy statements, press briefings, Executive Summaries, attributed minutes, and scenario-based presentations collectively help markets understand the Board's thinking. The goal is not precision, but clarity about how the central bank interprets risks and evaluates decisions in a least-regrets framework.

2. Transparency and Accountability as Foundations

Strategic communication supports transparency-vital for maintaining credibility in a small open economy. FPAS Mark II requires:

- clear explanations of decisions,
- structured presentation of scenarios,
- attribution of votes and individual reasoning, and
- timely publication of minutes.

Transparency allows the public to understand not just **what** the central bank decided but **why** the decision was made despite uncertainty. Accountability is reinforced when Board members explain their reasoning and interpretation of risks.

3. Scenario-Based Communication

FPAS Mark II organizes communication around scenarios rather than deterministic forecasts. Instead of projecting a single path for inflation and output, the central bank explains a distribution of possible outcomes-baseline, adverse, and optimistic.

Scenario-based communication shifts the narrative from "certainty projection" to "risk distribution." It helps prevent misinterpretation by showing that risks-not baselines- drive policy decisions. This strengthens credibility and allows markets to form more accurate expectations.

4. The Role of Expectations in Monetary Transmission

Monetary policy operates largely through expectations. Expectations of:

- future interest rates,
- inflation,
- financial conditions, and
- external developments

shape economic behaviour. FPAS Mark II acknowledges that clear communication reduces risk premiums, anchors expectations, and supports smoother transmission.

When the public understands the central bank's reaction to risks, volatility decreases and policy becomes more effective.

5. Communication Vehicles Under FPAS Mark II

The Central Bank of Armenia employs a suite of communication tools, each serving a specific purpose:

- Monetary Policy Report (MPR)

- Executive Statement
- Attributed minutes
- Press conferences
- Analyst briefings
- Media interviews
- Public speeches
- Educational materials via the Learning Hub

The MPR provides analytical depth; the Executive Statement summarizes decisions in plain language. Attributed minutes enhance transparency by documenting individual reasoning. Educational materials help build broader economic understanding in society.

6. Risk Communication and Credibility

FPAS Mark II emphasizes open and honest communication of risks. Concealing uncertainty damages credibility; acknowledging uncertainty strengthens it. The central bank gains trust by explaining what it knows, what it does not know, and how it plans to respond to different scenarios.

By framing decisions through a least-regrets mindset, communication conveys that policy is designed to minimize costly errors in an uncertain environment. This reinforces long-term credibility and stabilizes expectations.

Conclusion

Strategic communication under FPAS Mark II is a core policy instrument. It aligns expectations, manages uncertainty, strengthens accountability, and enhances credibility. By presenting structured scenario-based communication, the Central Bank of Armenia ensures that markets, analysts, and the public understand the rationale behind decisions and the risks shaping the policy environment.

Communication is not an accessory-it is essential to modern monetary policy.

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Script

Hello everyone, I'm **Susanna Grigoryan**, an economist at the **Central Bank of Armenia** and a **Level One student at the Global Forecasting School**. In this session, we'll explore the role of strategic communication in modern central banking and how it has become a core component of the FPAS Mark II framework.

Strategic communication is no longer a supplementary activity—it is a central instrument of modern monetary policy. Clear, transparent, and timely communication enhances credibility, stabilizes expectations, and allows the public and financial markets to better understand the central bank's assessment of risks. Under FPAS Mark II, communication is treated as a policy tool in its own right—one that manages uncertainty, anchors expectations, and strengthens accountability.

In earlier monetary regimes, communication was often treated as an afterthought, primarily descriptive rather than analytical. FPAS Mark II elevates it to a full-fledged policy instrument. It requires that communication not only describe what the central bank has done but explain how it thinks. Policy statements, press briefings, executive summaries, attributed minutes, and scenario-based explanations all work together to guide markets' understanding of policy intentions. The goal is not to provide perfect predictions but to clarify how the Board views risks and how it responds to uncertainty.

Strategic communication also strengthens transparency, which is essential for maintaining credibility—particularly in a small open economy like Armenia. FPAS Mark II sets high transparency standards by requiring clear explanations of policy decisions, structured presentation of scenarios, timely publication of minutes, and clear attribution of votes and reasoning. This transparency helps the public understand not only what the Bank decided but why it made those decisions in the face of uncertainty. Accountability is enhanced when Board members explain the reasoning behind their votes and their interpretations of risk.

Under FPAS Mark II, communication is structured around scenarios rather than deterministic forecasts. Instead of presenting a single baseline, the Bank explains multiple potential paths for inflation, growth, financial conditions, and external risks. This shifts communication from projecting certainty to presenting a distribution of risks. Scenario-based communication helps prevent misinterpretation by clarifying that it is risk—not a baseline—that drives policy. By explaining both upside and downside risks, the institution strengthens credibility and enables markets to form more informed expectations about future actions.

Monetary policy works largely through expectations—expectations of future interest rates, inflation, financial conditions, and global developments. FPAS Mark II recognizes that clear communication helps reduce uncertainty premiums and supports more efficient transmission of policy. When the public understands how the Bank interprets risks, how scenarios might evolve, and how credibility may influence inflation dynamics, expectations become more stable and better anchored. This reduces volatility and improves the effectiveness of policy interventions.

The Central Bank of Armenia uses a wide range of communication tools to reach different audiences. The Monetary Policy Report provides analytical depth. The Executive Statement offers a concise summary of decisions. Attributed minutes promote transparency. Press conferences and analyst briefings allow interactive clarification. Media interviews and public speeches extend outreach. And educational materials—such as those produced here at the CBA Learning Hub—help train future economists and increase public understanding of policy. Each communication vehicle serves a distinct but complementary role, reinforcing the Bank's credibility through consistency and clarity.

FPAS Mark II also emphasizes open risk communication. Concealing uncertainty damages credibility; embracing uncertainty strengthens it. The Bank gains trust by explaining what it knows, what it does not know, and how it plans to respond under different scenarios. By framing decisions through a least-regrets mindset, communication conveys that policy is designed to minimize costly mistakes when the world behaves differently than expected. This approach reinforces long-term credibility and helps maintain stable expectations even in times of volatility.

Strategic communication under FPAS Mark II is therefore central to effective monetary policy. It aligns expectations, manages uncertainty, strengthens accountability, and enhances credibility. By providing structured, scenario-based communication, the Central Bank of Armenia ensures that markets, analysts, and the public understand both the rationale behind decisions and the risks shaping them. Communication is no longer an accessory—it is a core component of modern monetary policy and one of the most powerful tools a central bank has to sustain trust.

Thank you for watching, and I would like to acknowledge **Douglas Laxton, Martin Galstyan, Vahe Avagyan, Jared Laxton, Asya Kostanyan, and Sopio Mkervalidze** at the **Global Forecasting School** and **The Better Policy Project** for their guidance and support in producing this video.

! Video 17 - Measuring and Enhancing Transparency

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November 2025

Note

In this session, we explore how FPAS Mark II strengthens transparency as a core element of credible monetary policy. Transparency under FPAS Mark II helps anchor expectations, reduce uncertainty, and support clearer policy transmission. It lowers the risk of misinterpretation and helps markets internalize the central bank's risk assessment rather than relying on speculation.

Under FPAS Mark II, transparency is embedded throughout the framework: in scenario design, communication, minutes, forecasts, and post-round assessments. It is not cosmetic. It is foundational to how we explain our decisions and how we earn and maintain credibility over time.

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Script

Hello, my name is Susanna Grigoryan. I'm a Level One student at the Global Forecasting School and an economist at the Central Bank of Armenia.

In this video, we explore how FPAS Mark II strengthens transparency as a core element of credible monetary policy.

Transparency under FPAS Mark II helps anchor expectations, reduce uncertainty, and support clearer policy transmission. Transparency lowers uncertainty, anchors expectations, and strengthens policy transmission. It reduces the risk of misinterpretation and helps markets internalize the central bank's risk assessment rather than relying on speculation. Under FPAS Mark II, transparency is integrated into scenario design, communication, minutes, forecasts, and post-round assessments.

Transparency has several dimensions. Procedural transparency explains how decisions are made. Analytical transparency explains the models, scenarios, and evidence. Outcome transparency presents policy results clearly. Accountability transparency attributes reasoning and votes. These dimensions ensure that the public understands not just the decision, but the framework and reasoning behind it.

The Central Bank of Armenia uses multiple transparency tools. These include attributed minutes, individual votes and justifications, the Executive Statement, the Monetary Policy Report, scenario-based communication, press conferences, and analyst briefings. These tools help clarify the Board's reasoning under uncertainty.

Scenario-based communication enhances transparency by showing alternative paths, risks, and trade-offs. It shifts expectations away from single deterministic forecasts and toward a balanced understanding of the distribution of risks. This strengthens public trust and policy credibility.

By explaining what we know, what we do not know, and how uncertainty affects decisions, we strengthen long-term credibility. Transparency ensures the public understands the reasoning behind decisions and the risks shaping them. Transparency is not cosmetic—it is foundational. FPAS Mark II embeds transparency into its analytical processes, communication strategy, and institutional culture, ensuring that credibility is earned through openness, clarity, and disciplined reasoning.

I would like to thank **Douglas Laxton, Martin Galstyan, Jared Laxton, Asya Kostanyan**, and **Sopio Mkervalidze** for their guidance and support in helping me prepare this video.

Video 18 - Decision-Making Transparency and Individual Accountability

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November 2025

Note

Introduction

Decision-making transparency is a defining feature of FPAS Mark II. It strengthens credibility by showing how individual policymakers interpret risks, how they apply the least-regrets mindset, and how their votes shape policy outcomes.

For the Central Bank of Armenia, institutionalizing decision-making transparency and individual accountability marks a major step toward world-class monetary policy governance.

1. Why Individual Accountability Matters

Under FPAS Mark II, accountability applies not only to the institution but to individual Board members. Accountability builds trust because it allows the public to see:

- who supported which decision,
- why they supported it,
- how their reasoning fits within the FPAS Mark II framework.

Clear accountability incentivizes:

- disciplined analysis,
- transparent reasoning,
- coherence in communication.

It reinforces credibility by demonstrating that decisions are deliberate and aligned with the Bank's mandate.

2. Attributed Minutes and Individual Votes

One of FPAS Mark II's most important innovations is the publication of attributed minutes. Each Board member's vote is recorded along with a brief explanation of their reasoning.

This level of transparency is exceptional globally. It enhances:

- democratic legitimacy,
- public and market understanding,
- policy predictability,
- trust in the decision-making process.

Attributed minutes reveal not only the final decision but also the differing interpretations of risks and scenarios among Board members-giving insight into uncertainty behind the round.

3. Explaining Disagreement Under Uncertainty

FPAS Mark II treats disagreement not as a weakness but as a strength. When Board members interpret risks differently, their perspectives help refine the institution's understanding of uncertainty.

The framework encourages:

- reasoned disagreement,
- openness to alternative scenarios,
- clarification of asymmetric risks,
- discipline in explaining judgments.

Individual accountability allows the public to see these differences without undermining the unity of the institution's final decision.

4. Publication Timelines and Transparency Discipline

FPAS Mark II includes strict timelines for publishing:

- the Executive Statement,
- attributed minutes,
- the Monetary Policy Report (MPR),
- supporting analytical materials.

Timeliness reinforces credibility. Delayed communication creates uncertainty and weakens policy transmission. Structured timing ensures consistent, predictable information without room for speculation.

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5. Preventing Discretion Through Openness

Individual accountability limits arbitrary decision-making. When Board members know their reasoning will be public, discussions become more:

- disciplined,
- evidence-based,
- aligned with FPAS Mark II,
- resistant to political pressure and biases.

Transparency reduces:

political pressure,

hidden biases,

excessive reliance on unstructured judgment,

discretionary deviations from analysis.

It creates a culture where decisions are grounded in coherent, scenario-based reasoning.

Conclusion

Decision-making transparency and individual accountability form a cornerstone of FPAS Mark II. By publishing votes, explanations, and attributed minutes, the Central Bank of Armenia demonstrates its commitment to disciplined reasoning, openness, and credible policymaking under uncertainty.

This strengthens trust, anchors expectations, and enhances the effectiveness of monetary policy.

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Script

Hello, my name is Susanna Grigoryan. I'm a Level One student at the Global Forecasting School and an economist at the Central Bank of Armenia.

In this video, we explore how FPAS Mark II strengthens policy credibility through decision-making transparency and individual accountability.

Decision-making transparency is a defining feature of FPAS Mark II. It strengthens credibility by showing how individual policymakers interpret risks, how they apply the least-regrets mindset, and how their votes shape policy outcomes. For the Central Bank of Armenia, institutionalizing decision-making transparency and individual accountability marks a major step toward world-class standards in monetary policy governance.

Under FPAS Mark II, accountability is not limited to the institution—it extends to individual Board members. Accountability builds trust because it allows the public to see who supported what decision, why they supported it, and how their reasoning fits within the FPAS Mark II framework. Clear accountability incentivizes disciplined analysis, transparency in reasoning, and coherence in policy communication. It reinforces credibility by showing that decisions are made thoughtfully and consistently with the Bank's mandate.

One of the most important innovations in FPAS Mark II is the publication of one-pagers. Each Board member's vote is recorded along with a brief explanation of their reasoning. This level of transparency is exceptional globally. It enhances democratic legitimacy, public and market understanding, policy predictability, and trust in the decision-making process. Attributed minutes reveal not only the final decision, but the differing interpretations of risks and scenarios among Board members—providing insight into the uncertainty behind each round.

FPAS Mark II treats disagreement not as a weakness but as a strength. When Board members interpret risks differently, their perspectives help refine the institution's understanding of uncertainty. The framework encourages reasoned disagreement, openness to alternative scenarios, clarification of asymmetric risks, and discipline in explaining judgments. Individual accountability allows the public to see these differences without undermining the unity of the institution's final decision.

FPAS Mark II also includes strict timelines for publishing the Executive Statement, attributed minutes, the Monetary Policy Report, and supporting analytical materials. Timeliness reinforces credibility. Delayed communication creates uncertainty and weakens policy transmission. The structured timing ensures that the public receives information consistently and predictably, without room for speculation.

Individual accountability limits arbitrary decision-making. When Board members know their reasoning will be publicly available, policy discussions become more disciplined, evidence-based, and aligned with the FPAS Mark II framework. Transparency reduces political pressure, hidden biases, excessive reliance on unstructured judgment, and discretionary deviations from the analytical system. It promotes a culture in which decisions are supported by coherent reasoning and scenario-based analysis.

Decision-making transparency and individual accountability form a cornerstone of FPAS Mark II. By publishing votes, explanations, and minutes, the Central Bank of Armenia demonstrates its commitment to openness, disciplined reasoning, and credible policymaking under uncertainty. This strengthens trust, anchors expectations, and enhances the effectiveness of monetary policy.

I would like to thank **Douglas Laxton**, **Jared Laxton**, **Asya Kostanyan**, and **Sopio Mkervalidze** for their guidance and support in helping me prepare this video.

Video 19 - The Accountability Framework

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November 2025

Note

Introduction

Accountability is a defining feature of FPAS Mark II. It ensures that monetary policy decisions are transparent, disciplined, and aligned with the mandate of price stability. The Accountability Framework provides structure for how the Central Bank of Armenia evaluates policy outcomes, reviews internal processes, and communicates results to the public.

1. Purpose of the Accountability Framework

Accountability reinforces credibility by demonstrating that:

- decisions are evaluated against clear objectives,
- mistakes are acknowledged and addressed,
- processes improve over time.

Under FPAS Mark II, accountability is continuous, not episodic. It is built directly into the 29-day policy cycle and the institutional communication strategy.

2. Measuring Policy Outcomes Against Objectives

Policy success is assessed against:

- inflation developments relative to the target,
- measures of underlying inflation,
- economic slack and labour market indicators,
- financial stability metrics,
- external and geopolitical risks.

FPAS Mark II emphasizes evaluating outcomes within an uncertainty-aware, least-regrets framework-avoiding rigid scorekeeping while maintaining transparency.

3. Internal Review Mechanisms

Internal accountability is strengthened through:

- post-round reviews,
- forecast performance assessment,
- scenario effectiveness evaluations,
- credibility and expectations analysis,
- communication quality checks.

These reviews build institutional memory and reinforce learning across policy rounds.

4. External Review and Public Accountability

Transparency also requires external accountability. This includes:

- publication of the Monetary Policy Report,
- the Executive Statement,
- attributed minutes,
- press conferences,
- annual reviews.

External accountability reinforces democratic legitimacy and allows analysts, experts, and the public to scrutinize policy decisions under FPAS Mark II.

5. Embedding Accountability in Institutional Culture

Accountability is not merely procedural-it is cultural. FPAS Mark II promotes:

- humility in decision-making,
- openness about uncertainty,
- willingness to revise assessments,
- clear communication of risks.

This culture ensures that accountability becomes a source of strength rather than a constraint.

Conclusion

The Accountability Framework of FPAS Mark II ensures that the Central Bank of Armenia remains transparent, credible, and aligned with its mandate. By integrating accountability into analysis, communication, and institutional learning, the Bank strengthens public trust and enhances policy effectiveness.

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Script

Hello, my name is Susanna Grigoryan. I'm a Level One student at the Global Forecasting School and an economist at the Central Bank of Armenia.

In today's video, we explore how FPAS Mark II strengthens credibility through a robust Accountability Framework.

Accountability is a defining feature of FPAS Mark II. It ensures that monetary policy decisions are transparent, disciplined, and aligned with the mandate of price stability. The Accountability Framework gives structure to how the Central Bank of Armenia evaluates policy outcomes, reviews internal processes, and communicates results to the public.

Accountability reinforces credibility by demonstrating that decisions are evaluated against clear objectives, that mistakes are acknowledged and addressed, and that processes improve over time. Under FPAS Mark II, accountability is continuous, not

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episodic. It is built directly into the 29-day policy cycle and the institutional communication strategy.

Policy success is assessed against inflation developments relative to the target, measures of underlying inflation, economic slack and labour market indicators, financial stability metrics, and external and geopolitical risks. FPAS Mark II emphasizes evaluating outcomes within an uncertainty-aware, least-regrets framework, avoiding rigid scorekeeping while maintaining transparency.

Internal accountability is strengthened through post-round reviews, forecast performance assessment, scenario effectiveness evaluations, credibility and expectations analysis, and communication quality checks. These reviews build institutional memory and reinforce learning across policy rounds.

Transparency also requires external accountability. This includes the publication of the Monetary Policy Report, the Executive Statement, attributed minutes, press conferences, and annual reviews. External accountability reinforces democratic legitimacy and allows experts, analysts, and the public to scrutinize policy decisions under FPAS Mark II.

Accountability is not merely procedural—it is cultural. FPAS Mark II promotes humility in decision-making, openness about uncertainty, willingness to revise assessments, and clear communication of risks. This culture ensures that accountability becomes a source of strength rather than constraint.

The Accountability Framework of FPAS Mark II ensures that the Central Bank of Armenia remains transparent, credible, and aligned with its mandate. By integrating accountability into analysis, communication, and institutional learning, the Bank strengthens public trust and enhances policy effectiveness.

I would like to thank **Douglas Laxton**, **Jared Laxton**, **Asya Kostanyan**, and **Sopio Mkervalidze** for their guidance and support in helping me prepare this video.

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